

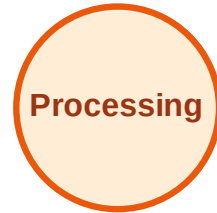
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

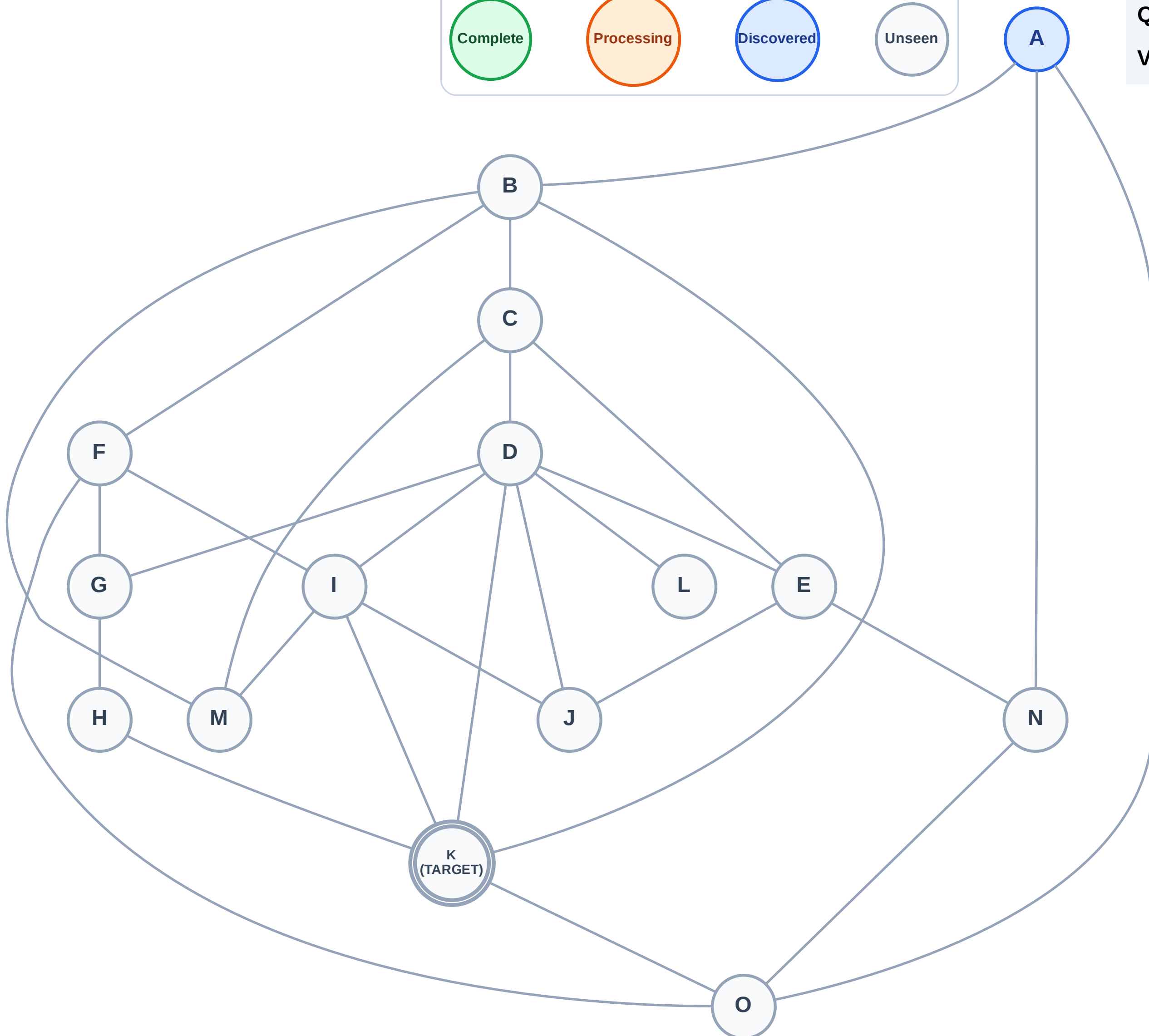


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

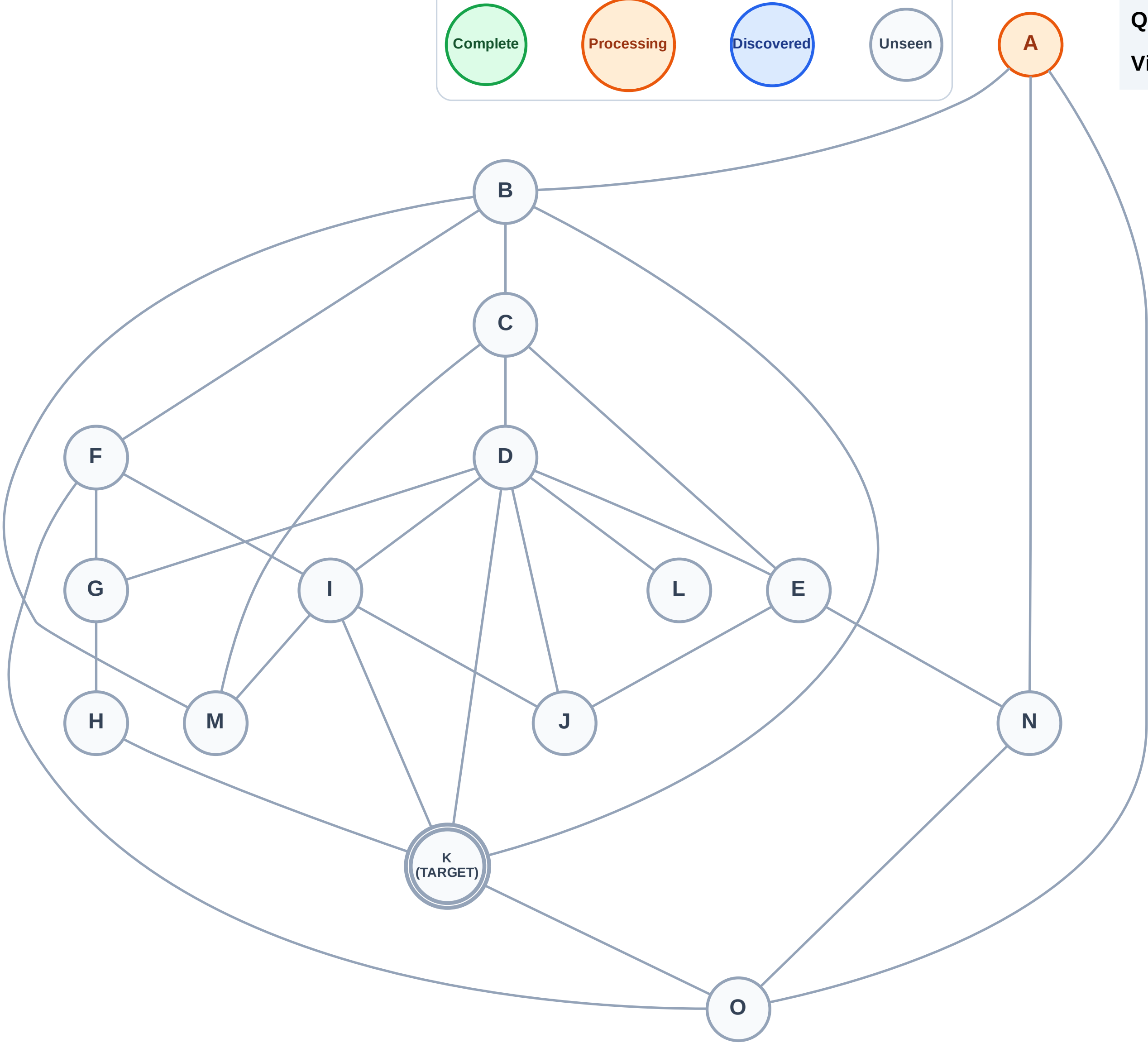
Complete

Processing

Discovered

Unseen

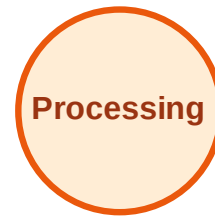
Queue: (empty)
Visited: (none)



BFS Traversal

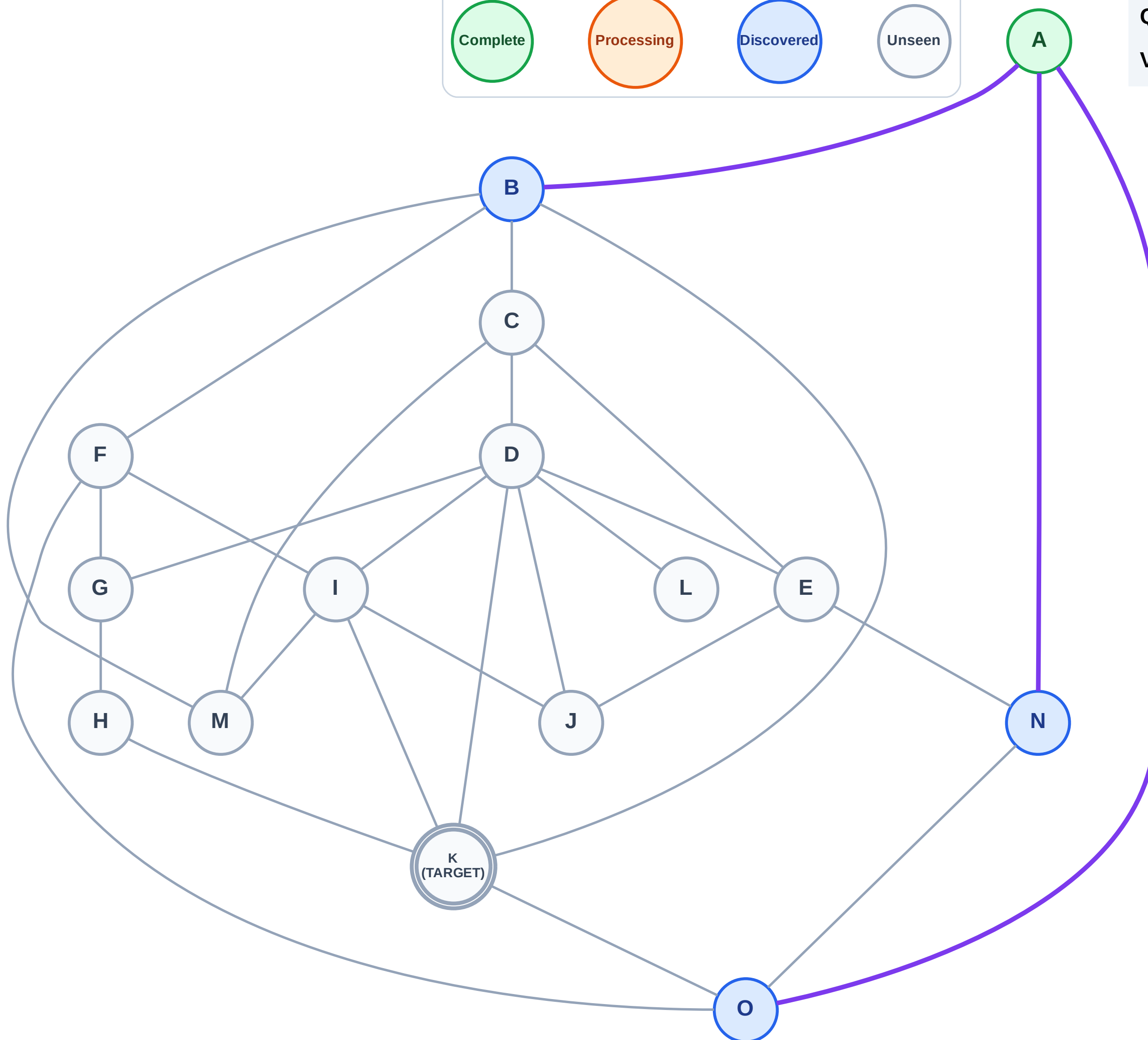
Step 3: Discover B, N, O from A

Legend



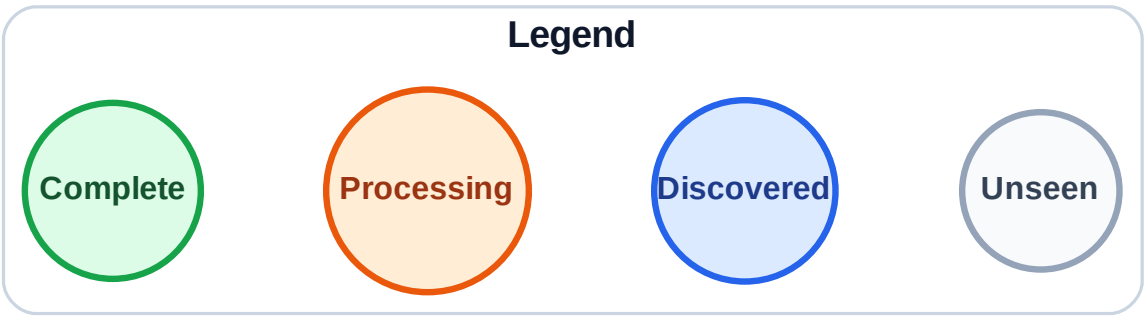
Queue: B → N → O

Visited: A



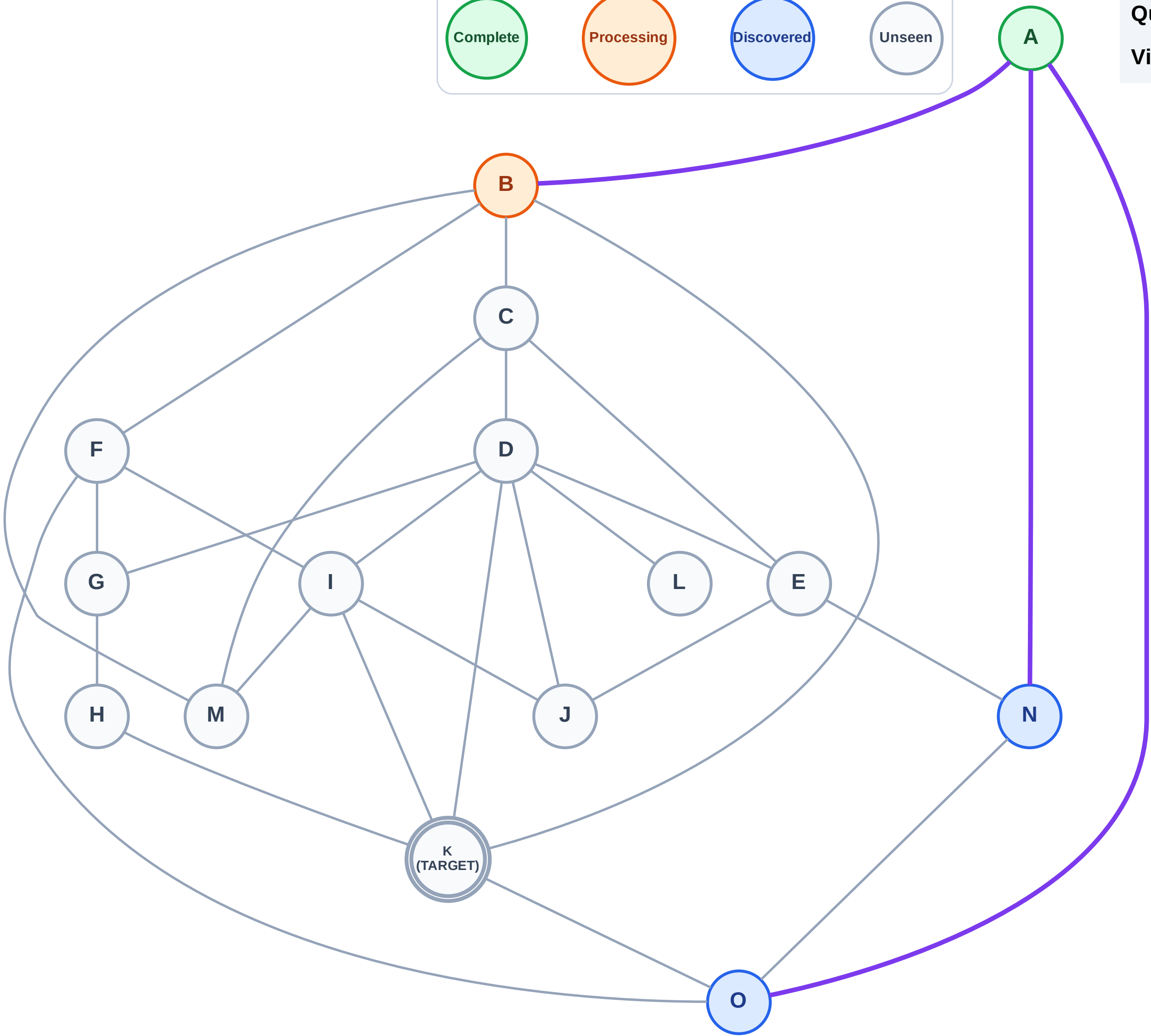
BFS Traversal

Step 4: Process node B



Queue: N → O

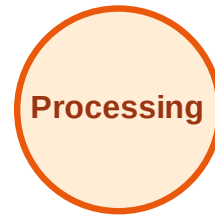
Visited: A



BFS Traversal

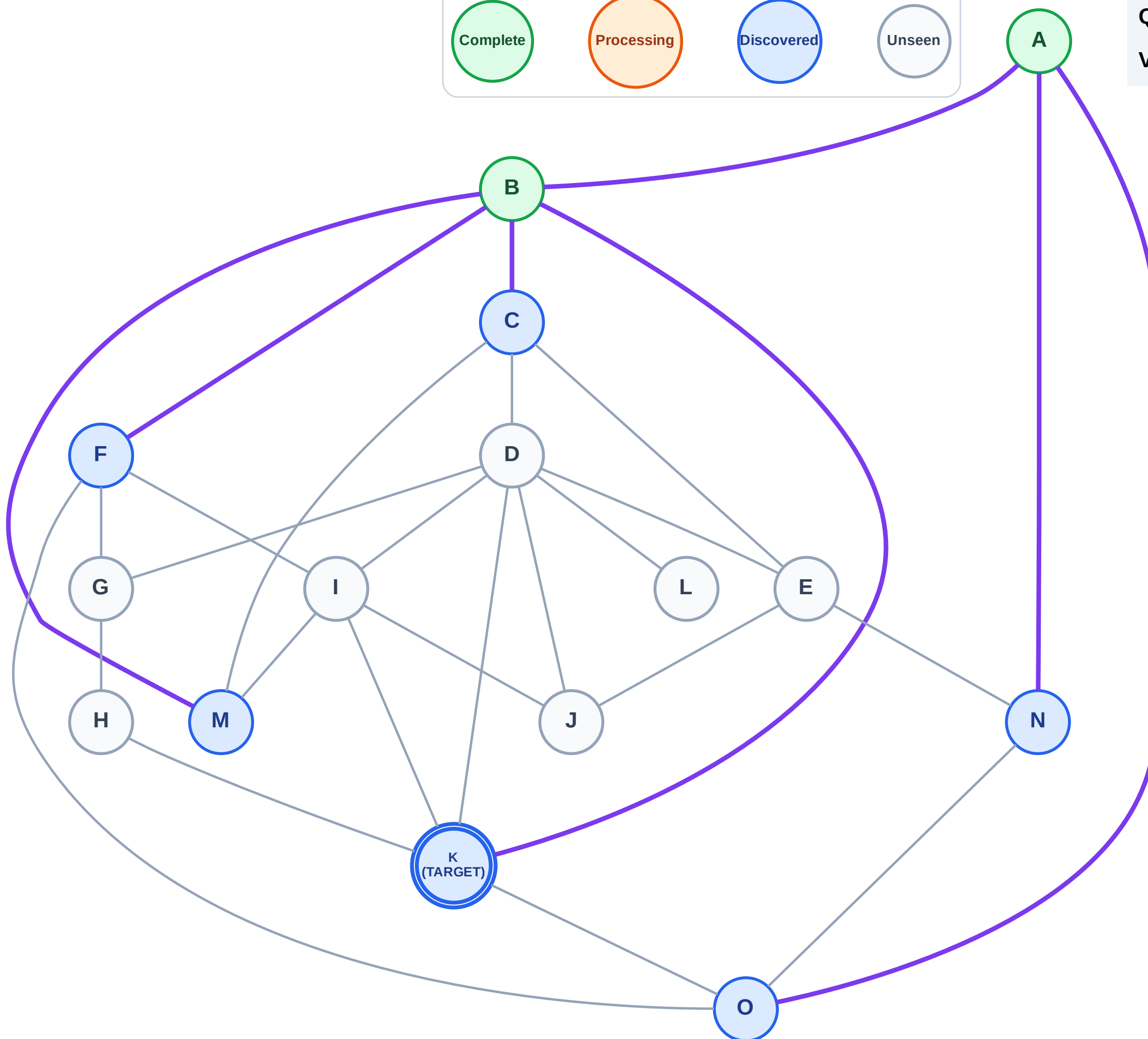
Step 5: Discover C, F, K, M from B

Legend



Queue: N → O → C → F → K → M

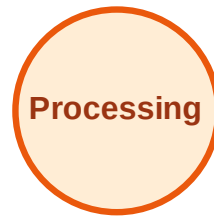
Visited: A, B



BFS Traversal

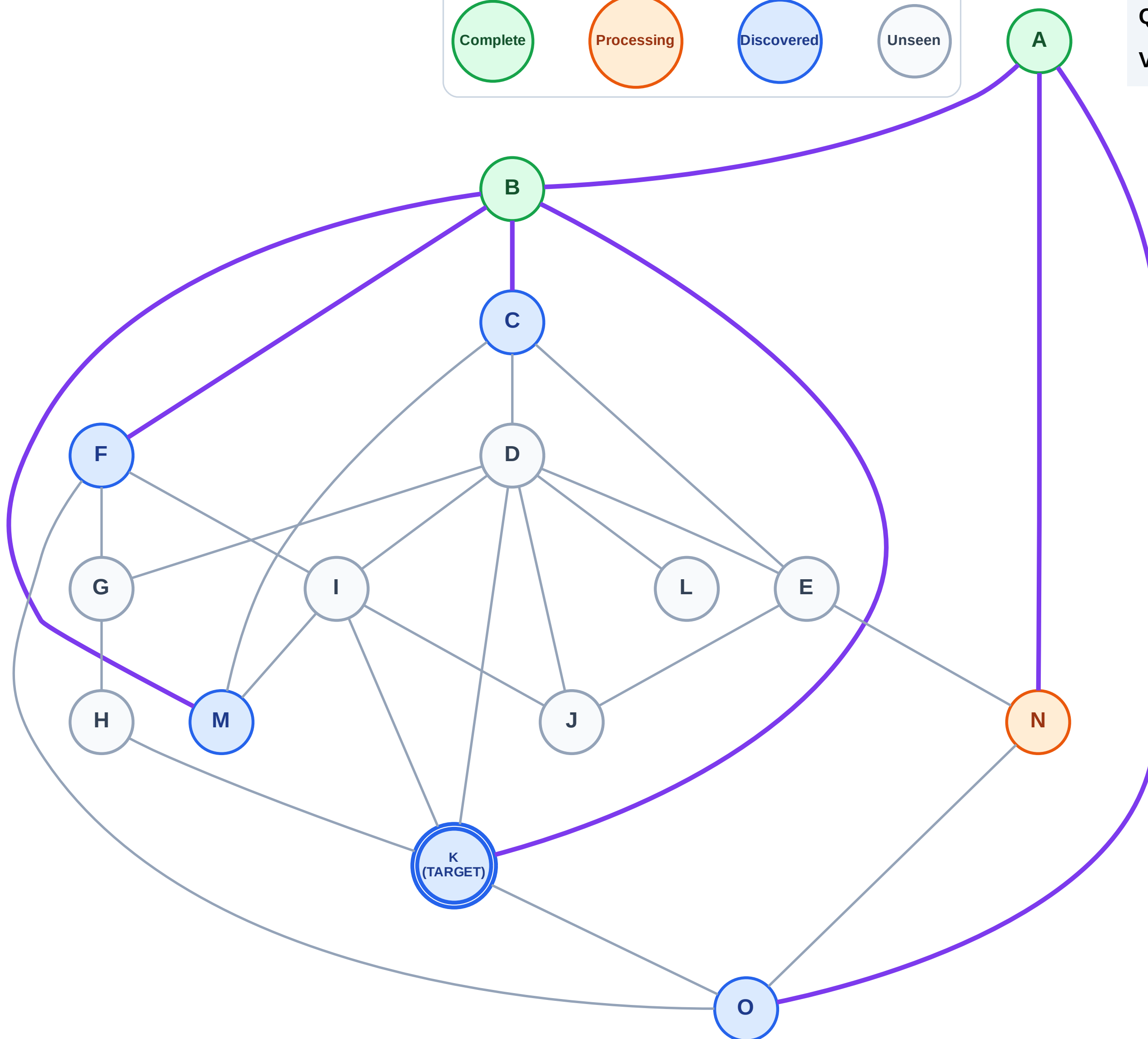
Step 6: Process node N

Legend



Queue: O → C → F → K → M

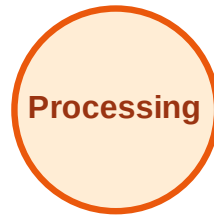
Visited: A, B



BFS Traversal

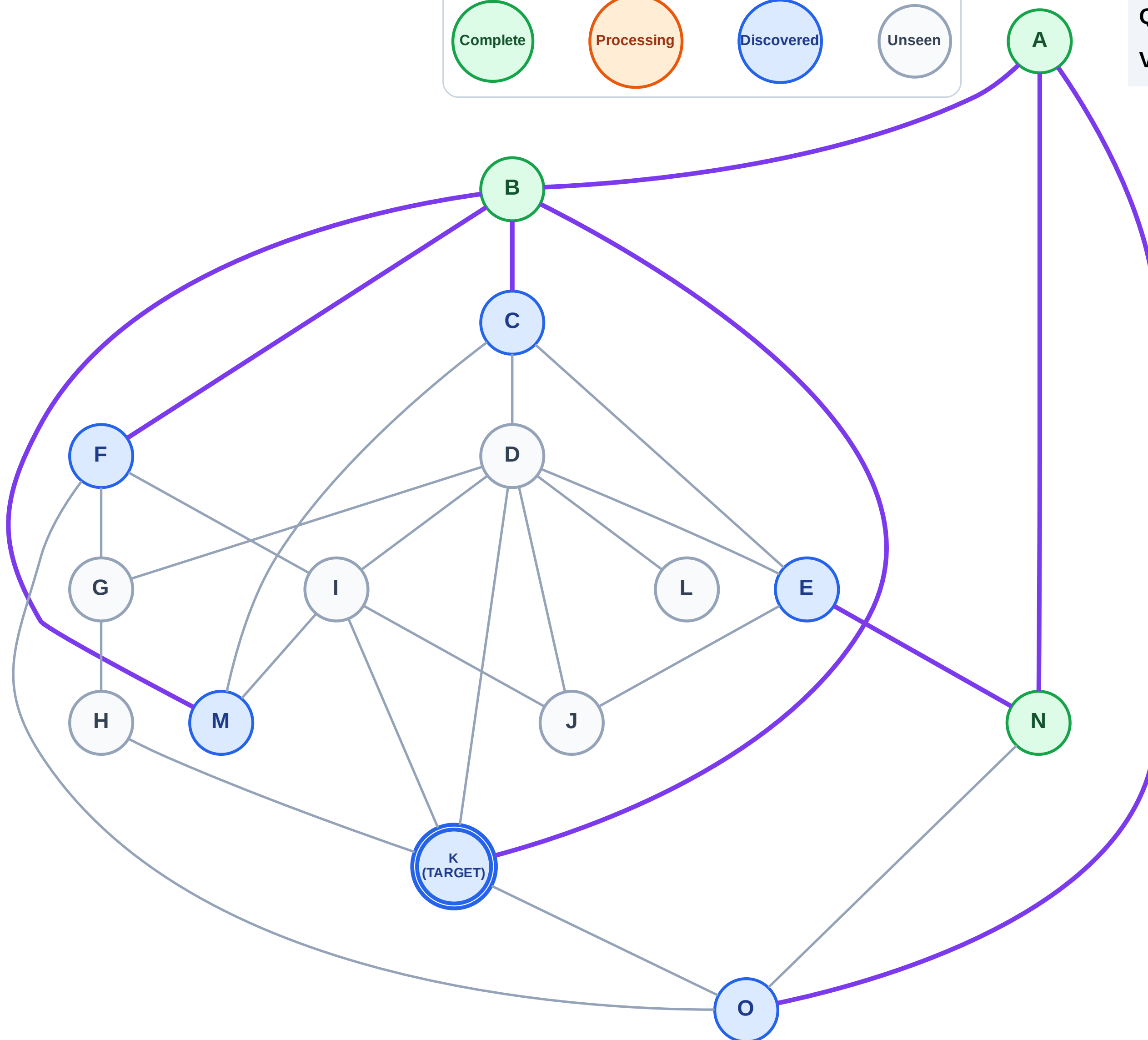
Step 7: Discover E from N

Legend



Queue: O → C → F → K → M → E

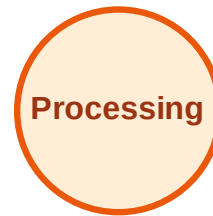
Visited: A, B, N



BFS Traversal

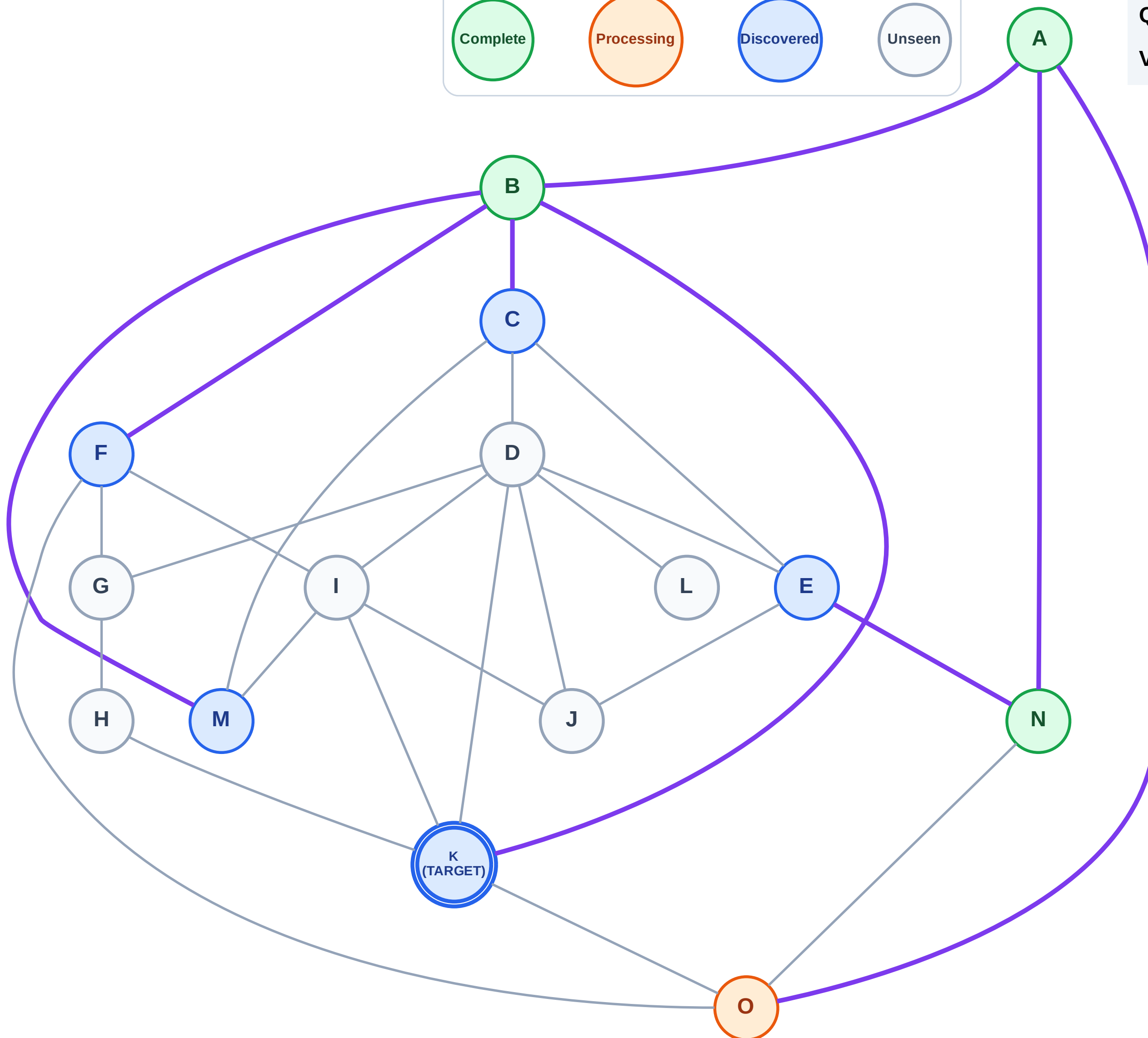
Step 8: Process node O

Legend



Queue: C → F → K → M → E

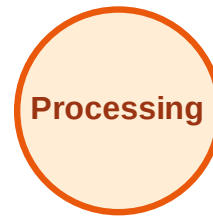
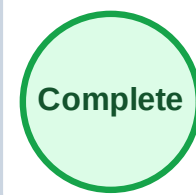
Visited: A, B, N



BFS Traversal

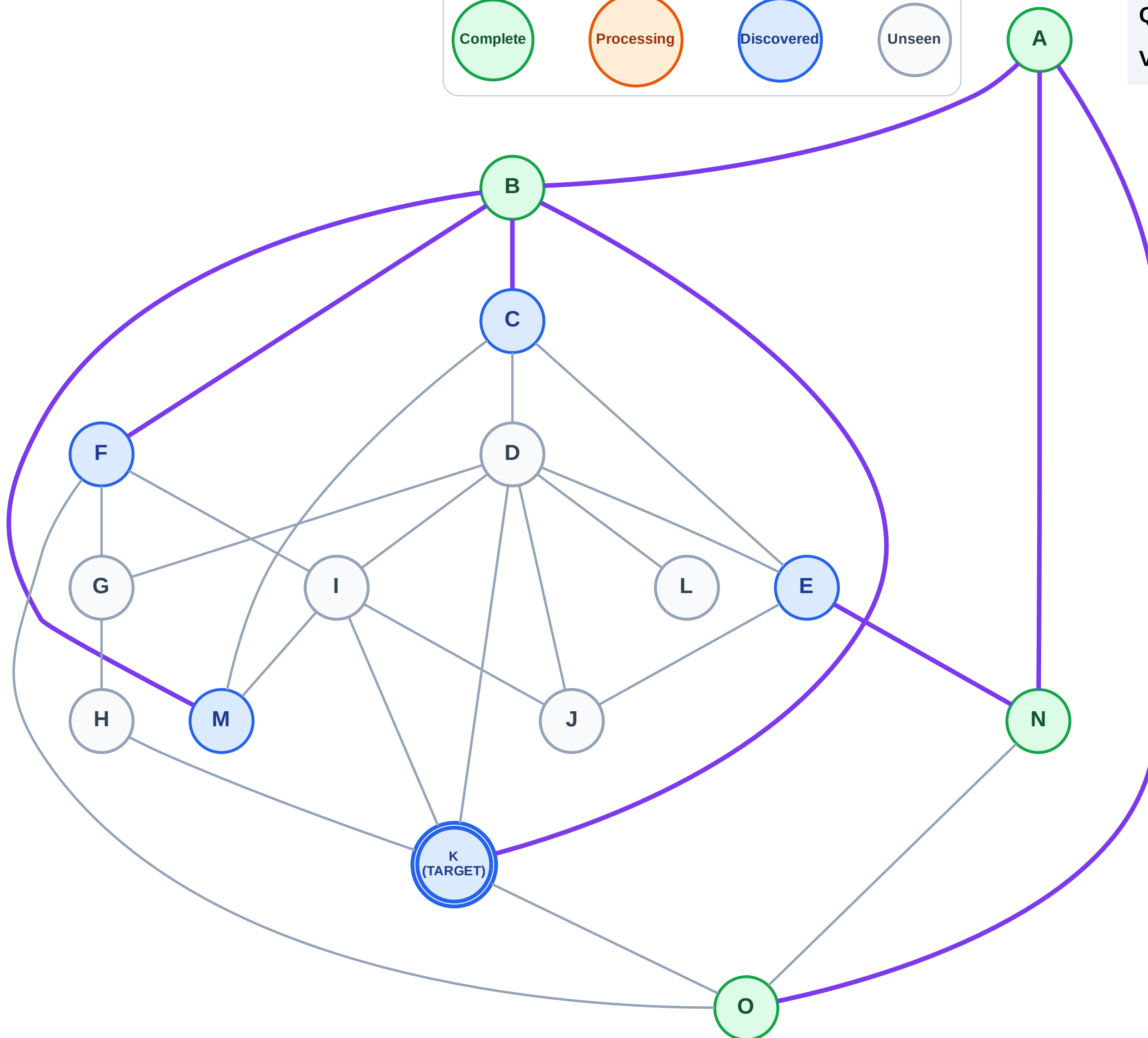
Step 9: No new neighbors from O

Legend



Queue: C → F → K → M → E

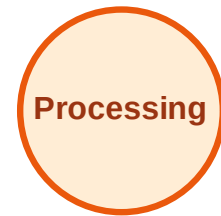
Visited: A, B, N, O



BFS Traversal

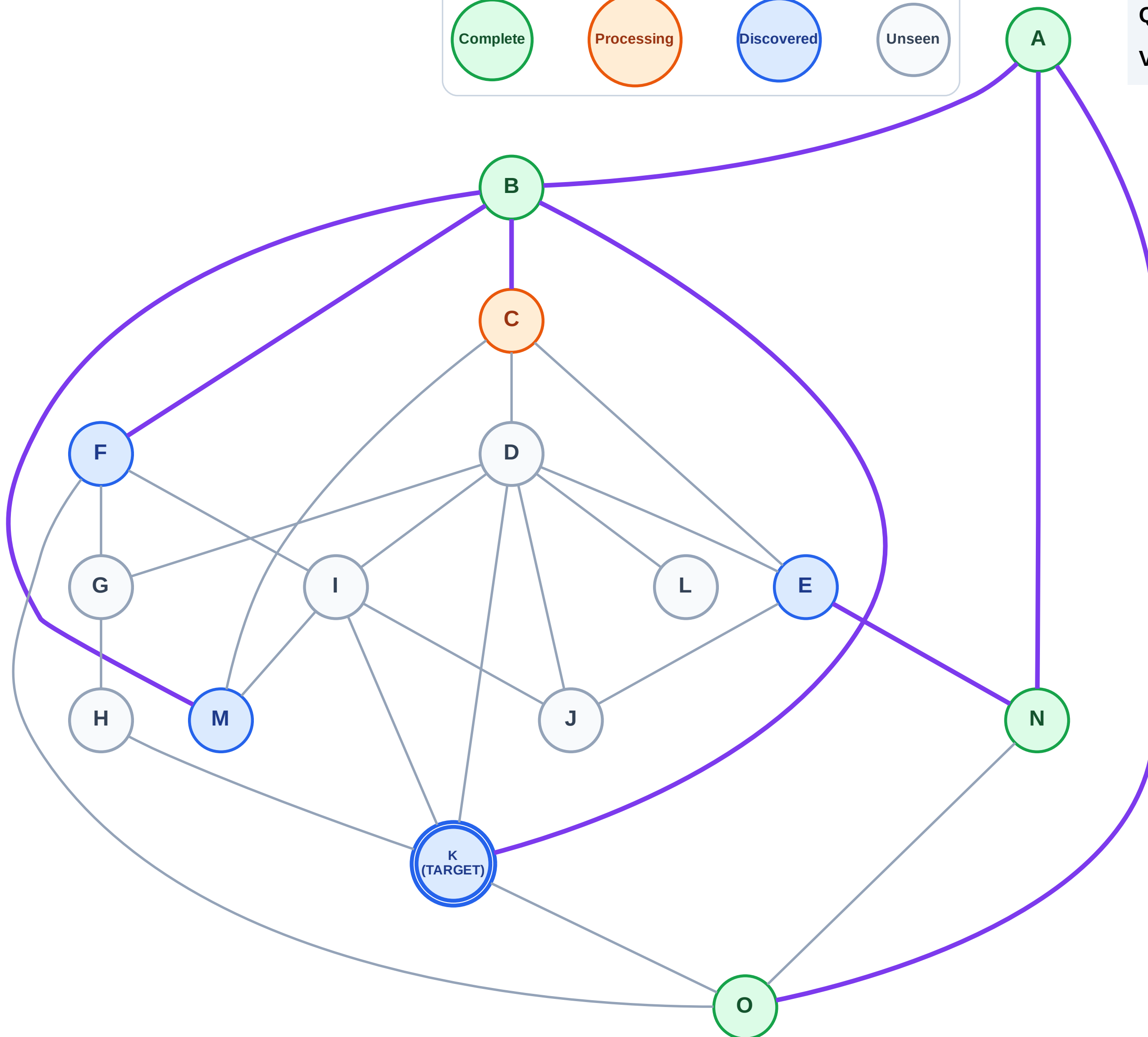
Step 10: Process node C

Legend



Queue: F → K → M → E

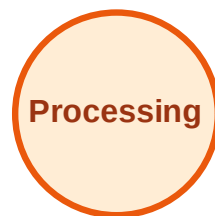
Visited: A, B, N, O



BFS Traversal

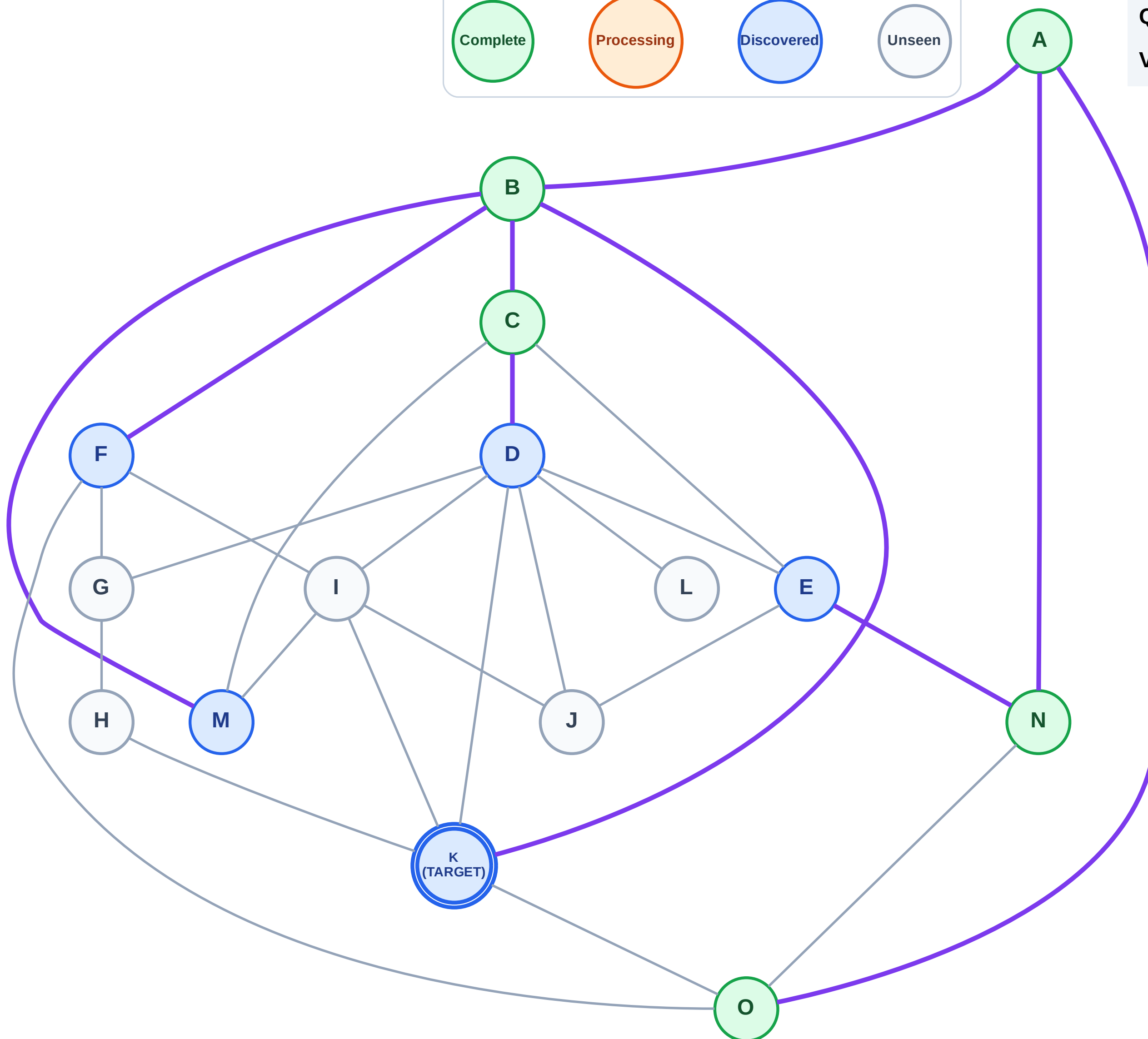
Step 11: Discover D from C

Legend



Queue: F → K → M → E → D

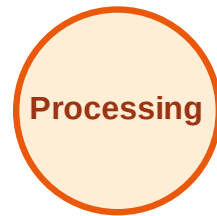
Visited: A, B, C, N, O



BFS Traversal

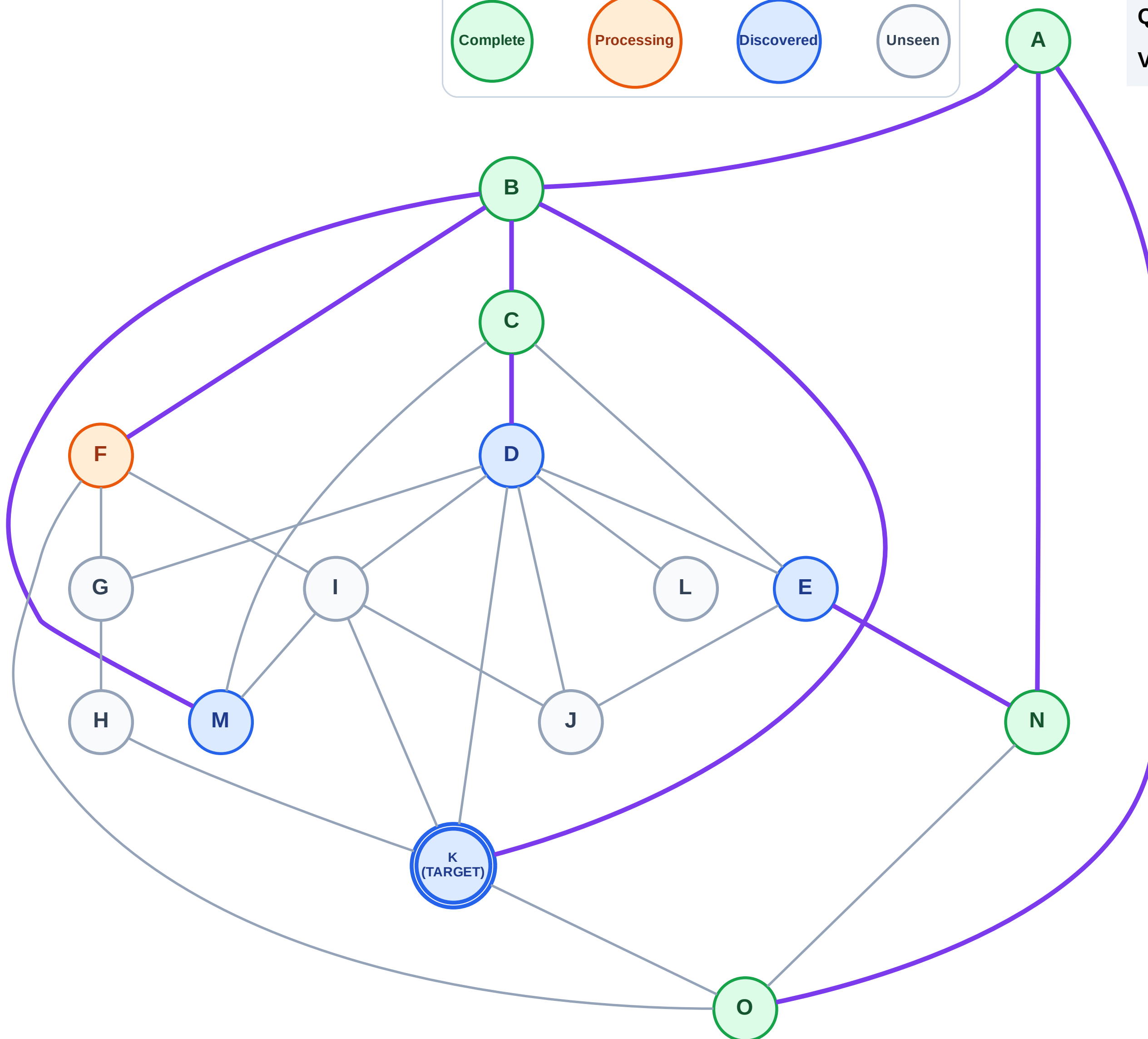
Step 12: Process node F

Legend



Queue: K → M → E → D

Visited: A, B, C, N, O



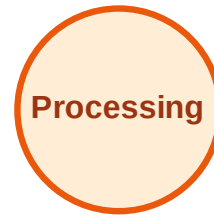
BFS Traversal

Step 13: Discover G, I from F

Legend



Complete



Processing



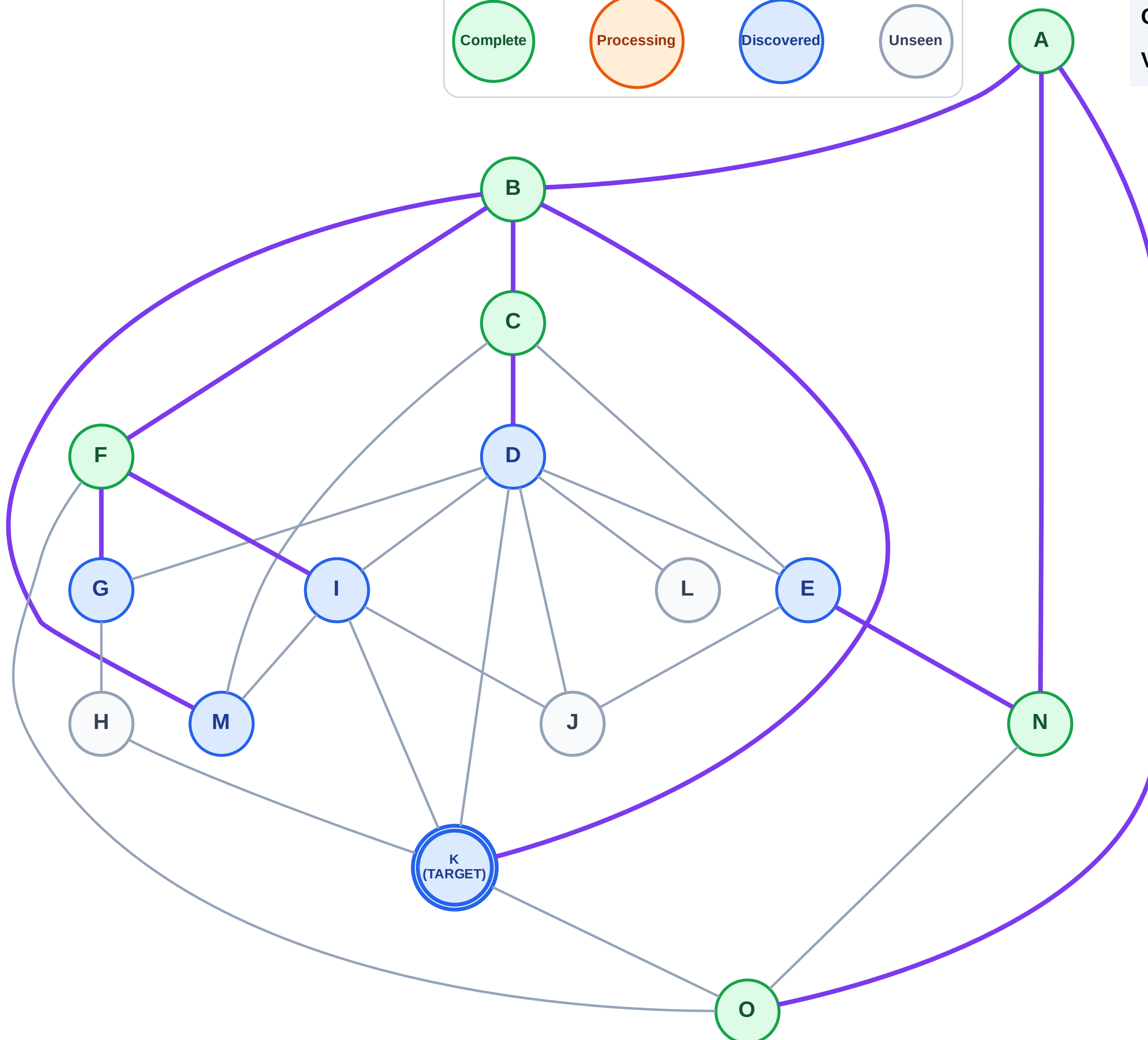
Discovered



Unseen

Queue: K → M → E → D → G → I

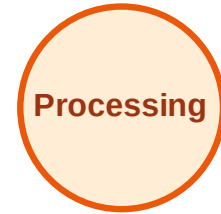
Visited: A, B, C, F, N, O



BFS Traversal

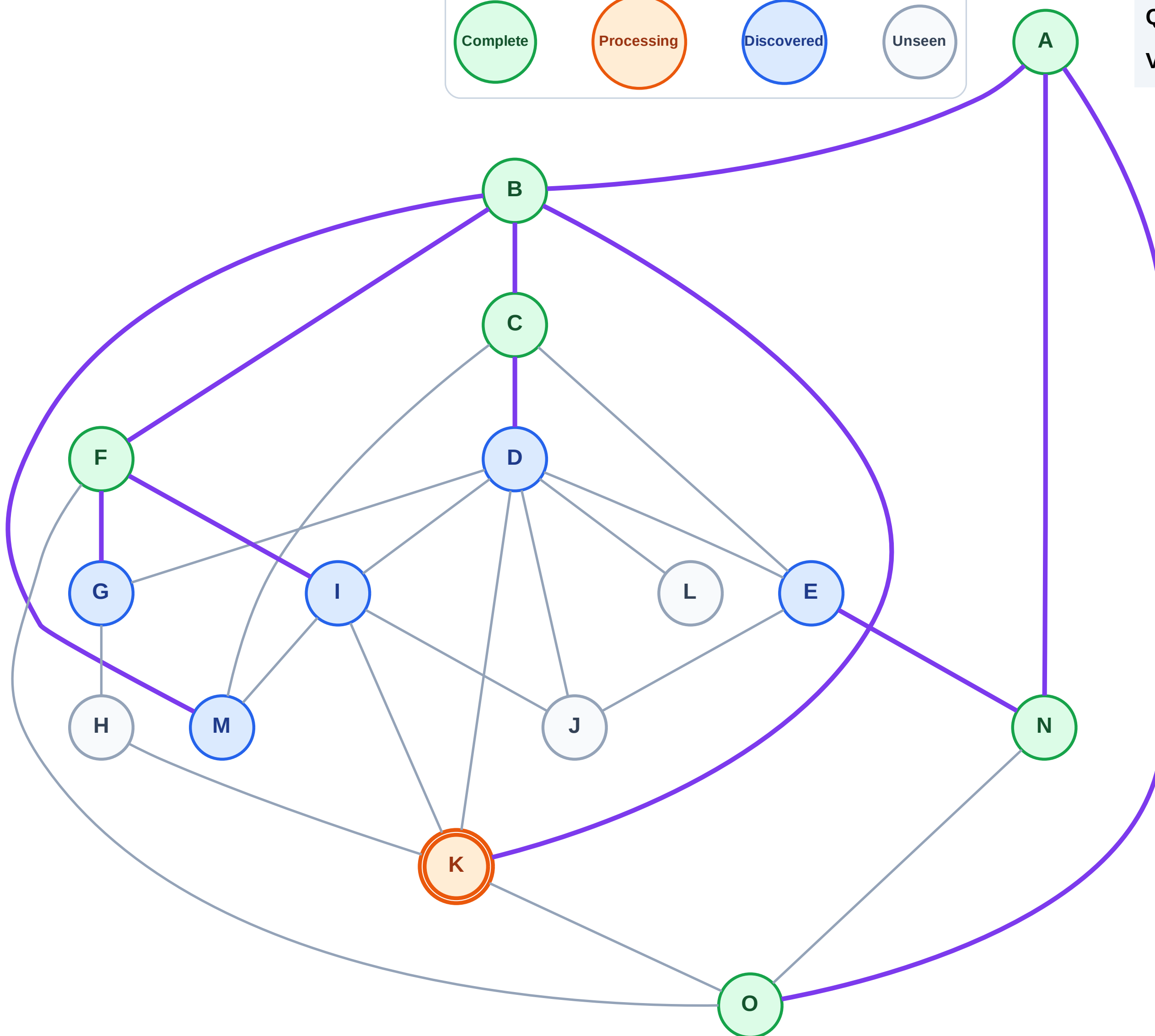
Step 14: Process node K

Legend



Queue: M → E → D → G → I

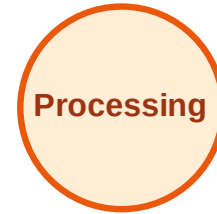
Visited: A, B, C, F, N, O



BFS Traversal

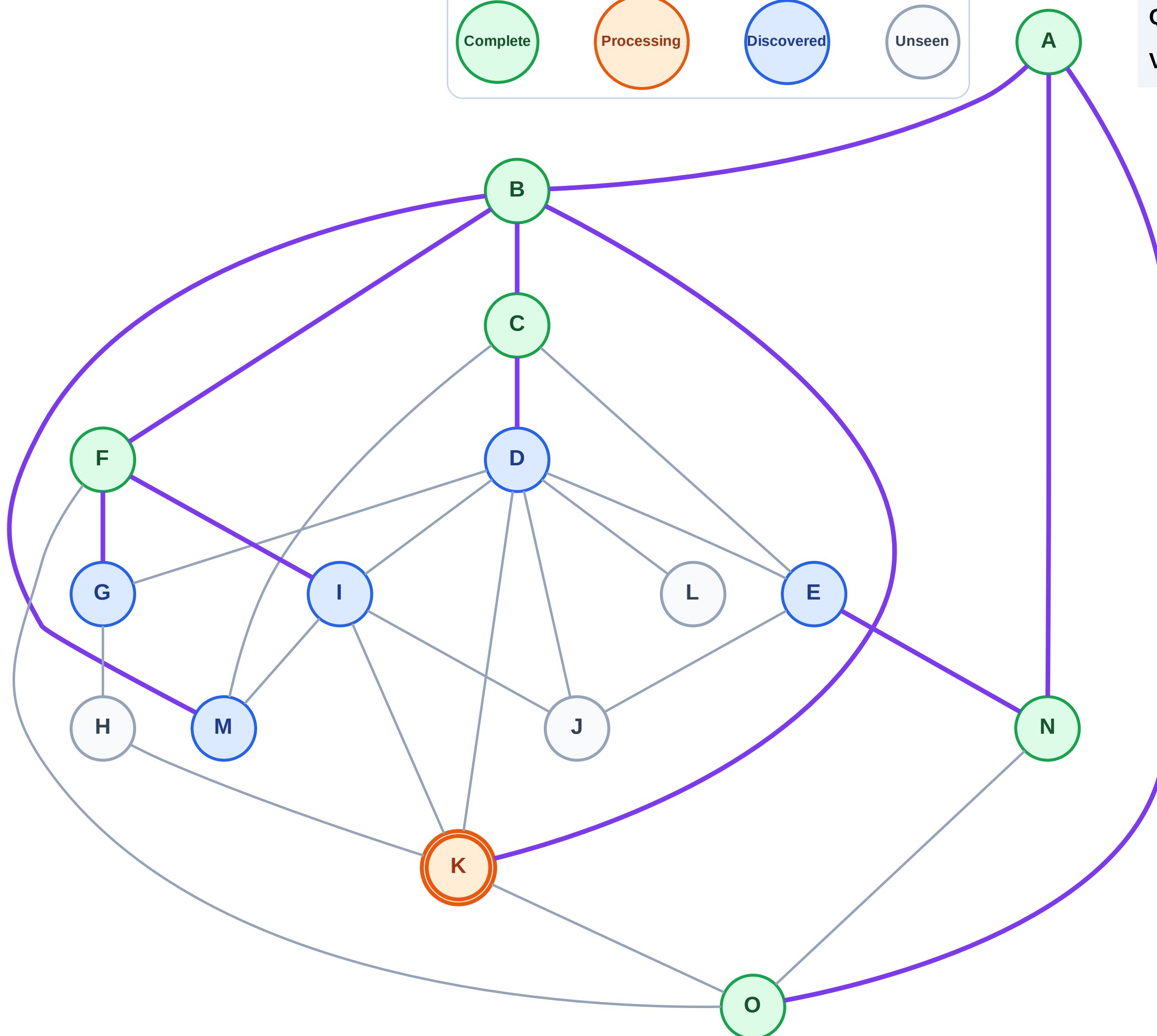
Step 15: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: M → E → D → G → I

Visited: A, B, C, F, K, N, O



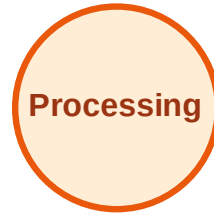
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

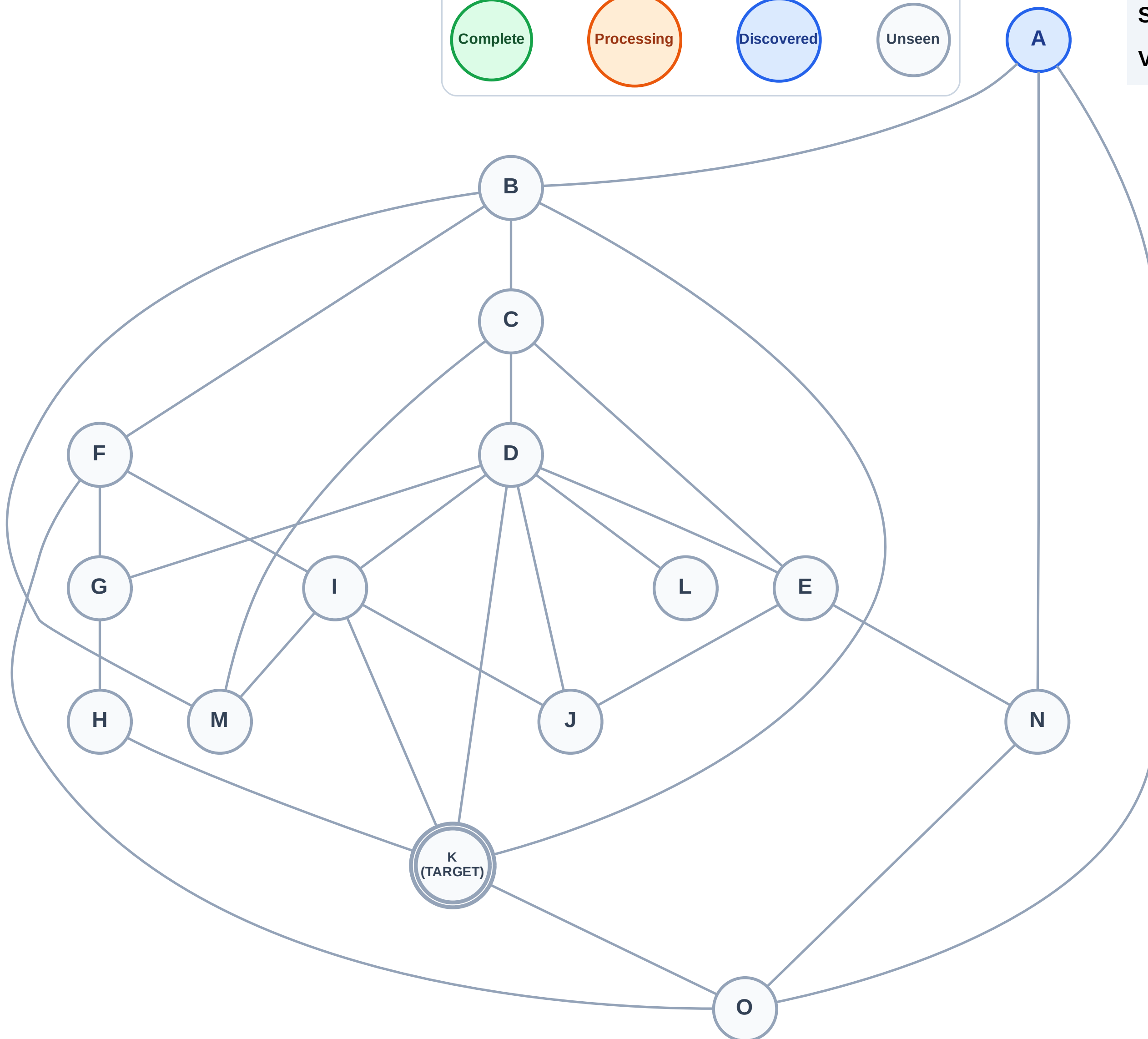


Discovered



Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

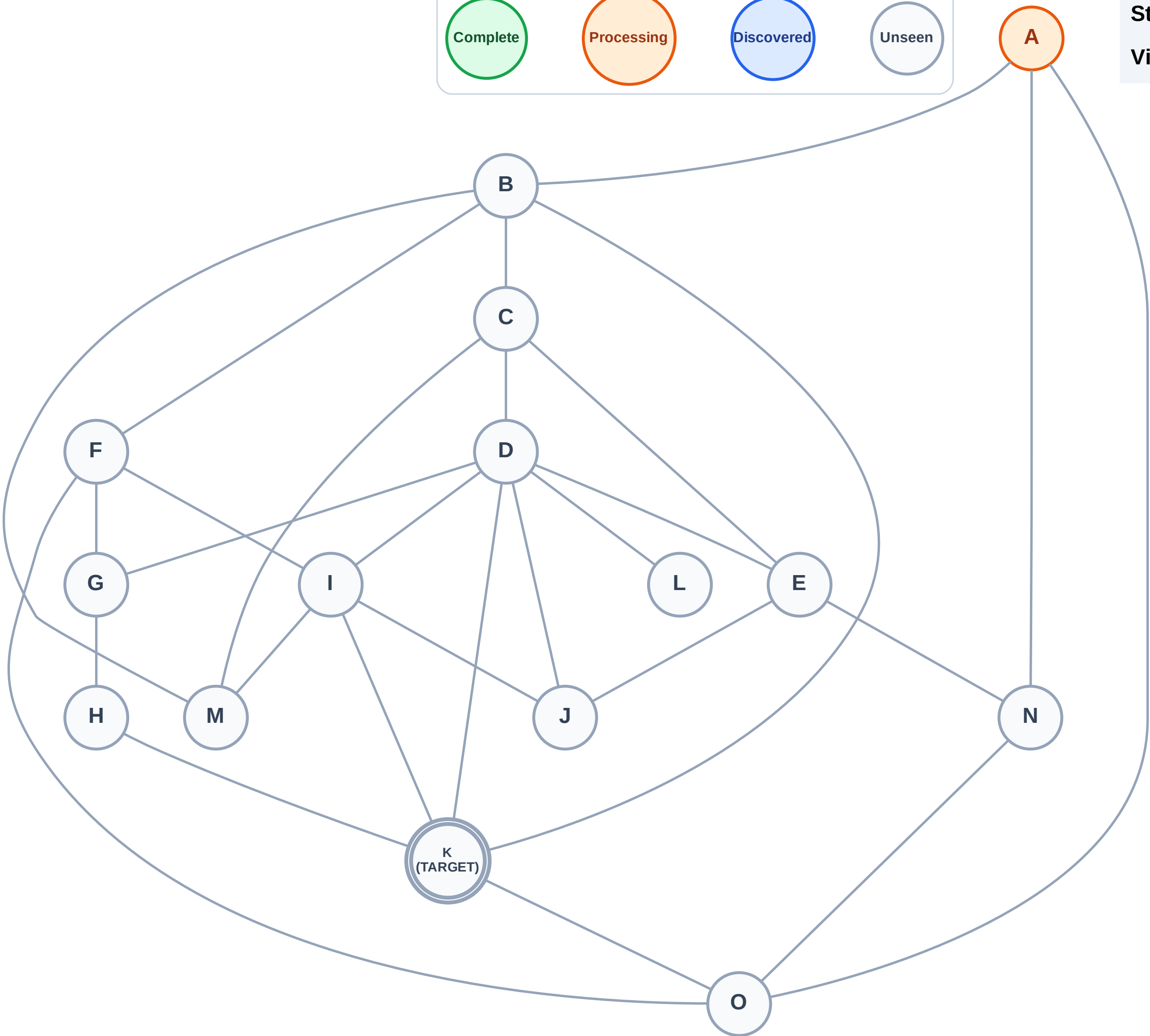
Step 2: Process node A

Legend



Stack (top at right): (empty)

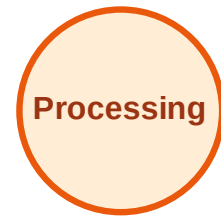
Visited: (none)



DFS Traversal

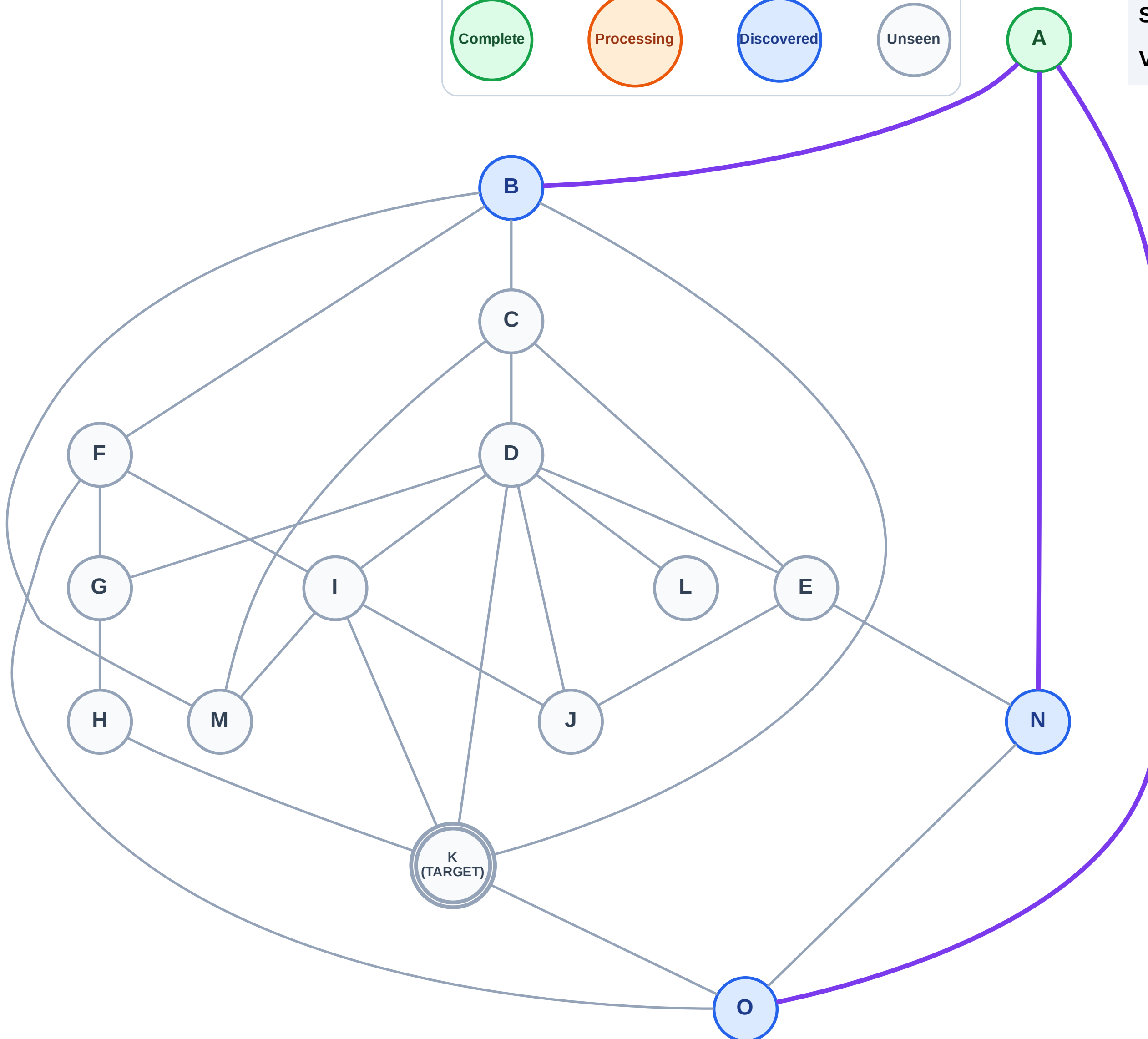
Step 3: Discover O, N, B from A

Legend



Stack (top at right): O | N | B

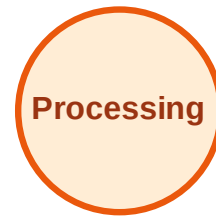
Visited: A



DFS Traversal

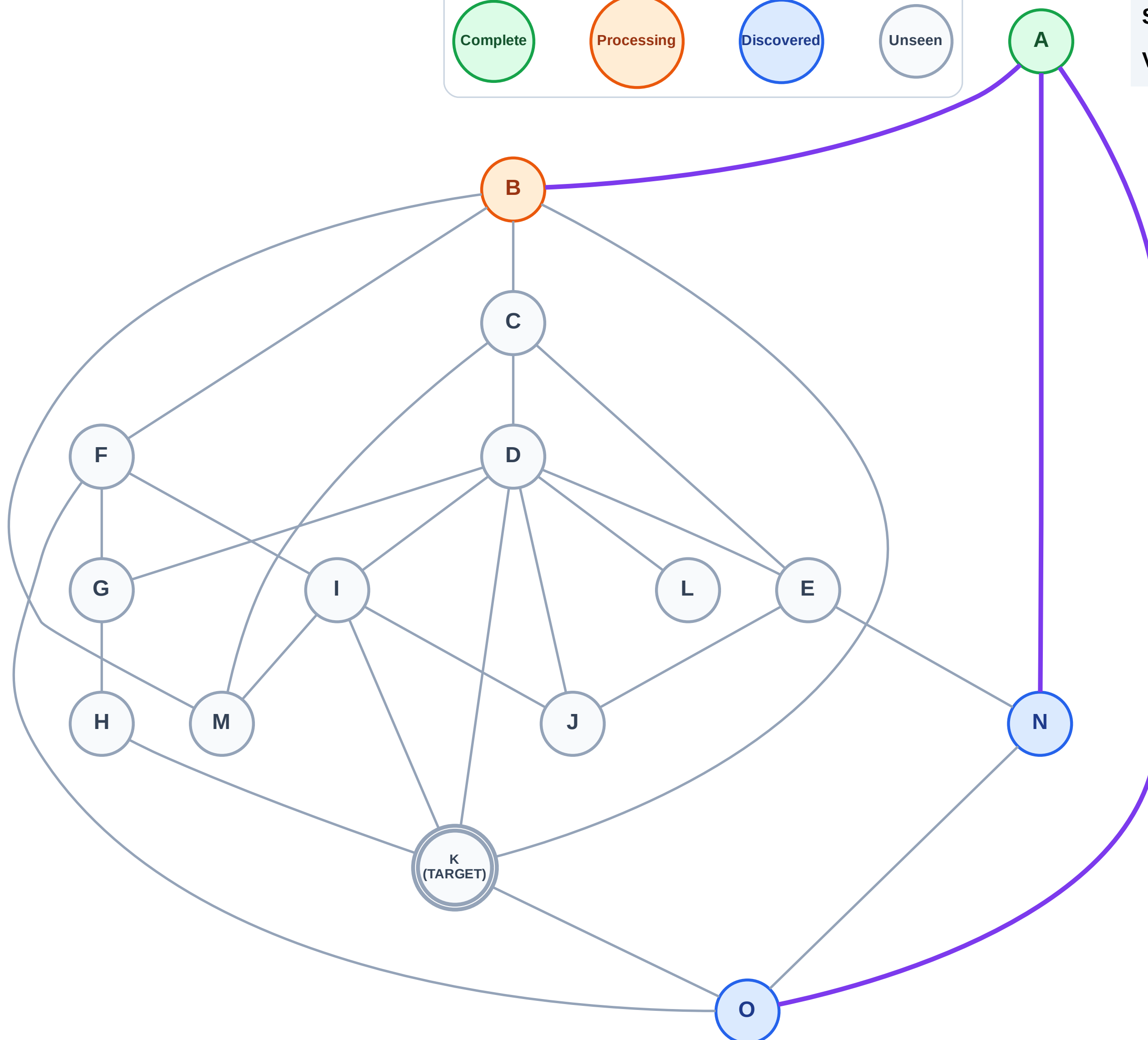
Step 4: Process node B

Legend



Stack (top at right): O | N

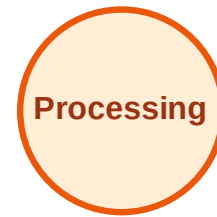
Visited: A



DFS Traversal

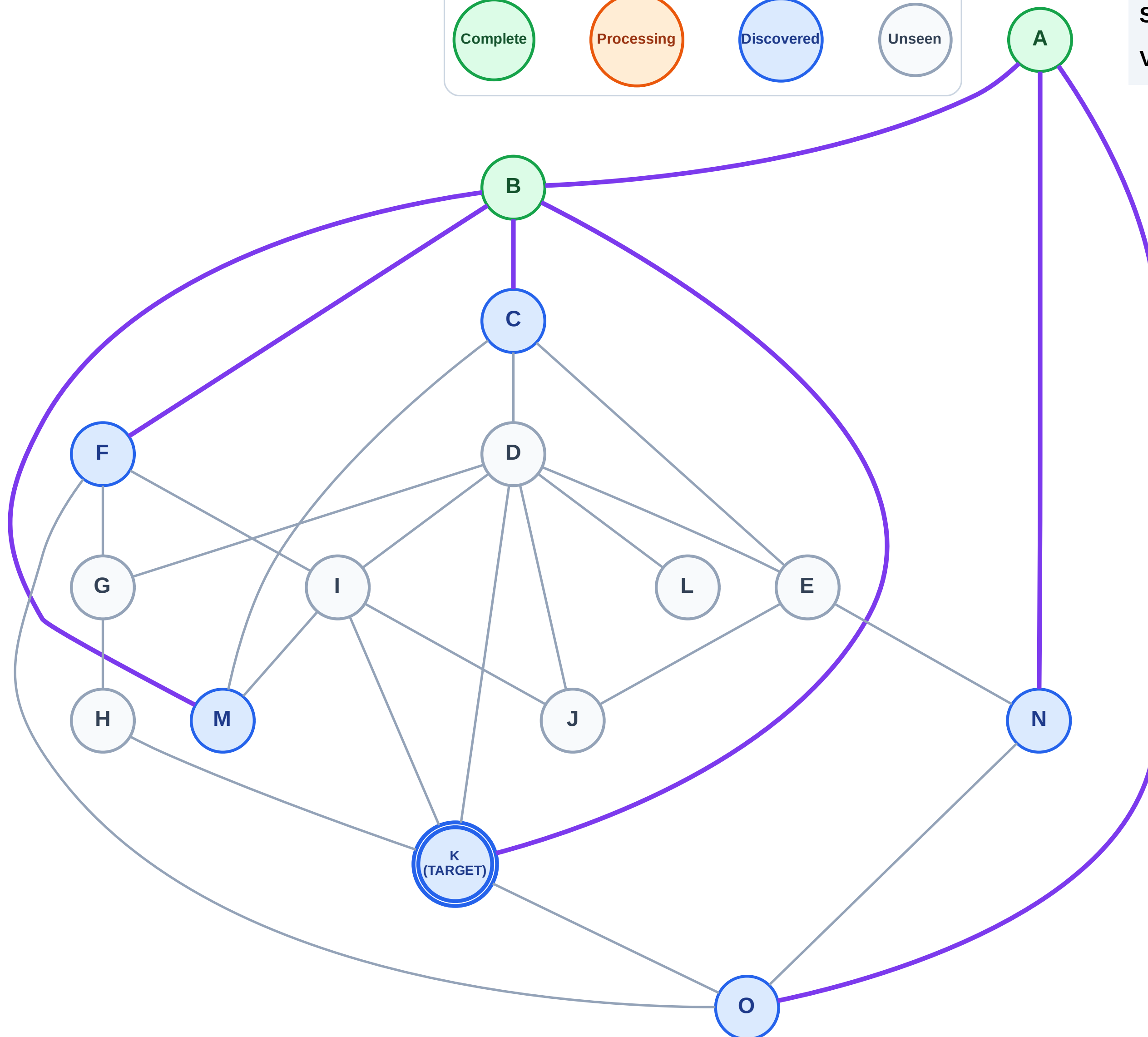
Step 5: Discover M, K, F, C from B

Legend



Stack (top at right): O | N | M | K | F | C

Visited: A, B



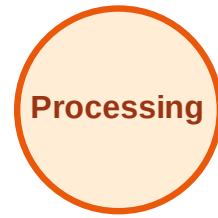
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



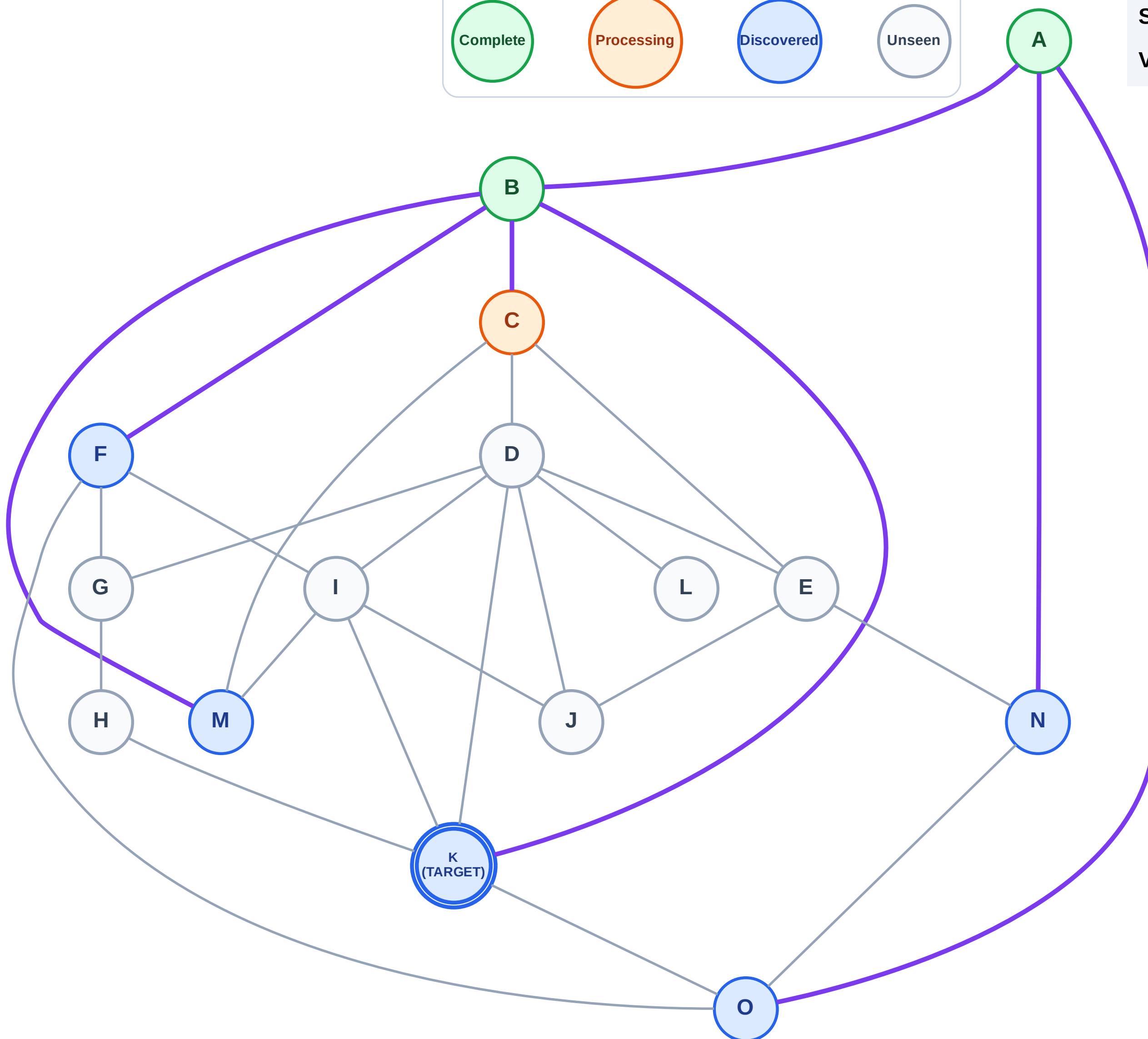
Discovered



Unseen

Stack (top at right): O | N | M | K | F

Visited: A, B



Step 7: Discover E, D from C

Step 7: Discover E, D from C

Legend

Complete

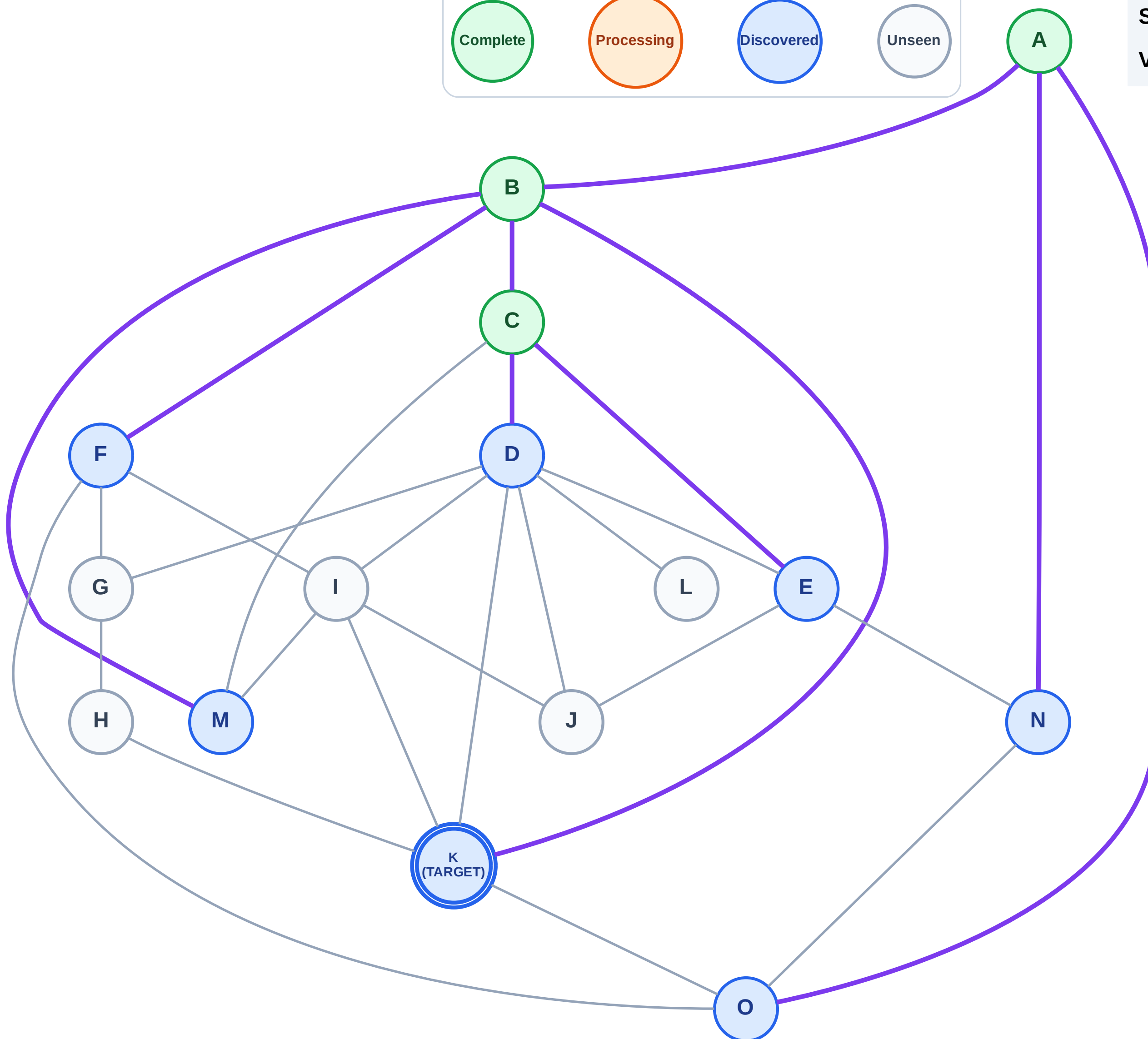
Processing

Discovered

Unseen

Stack (top at right): O | N | M | K | F | E | D

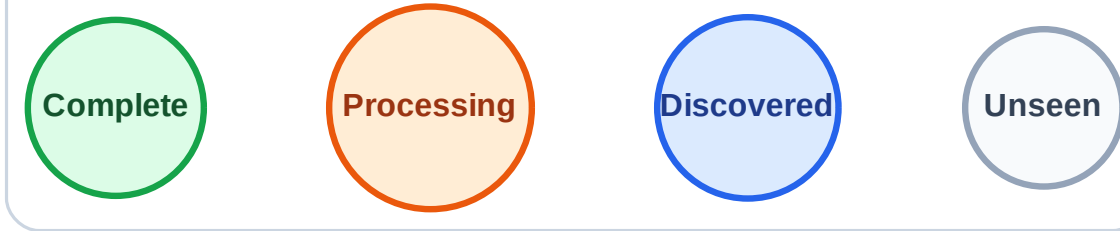
Visited: A, B, C



Step 8: Process node D

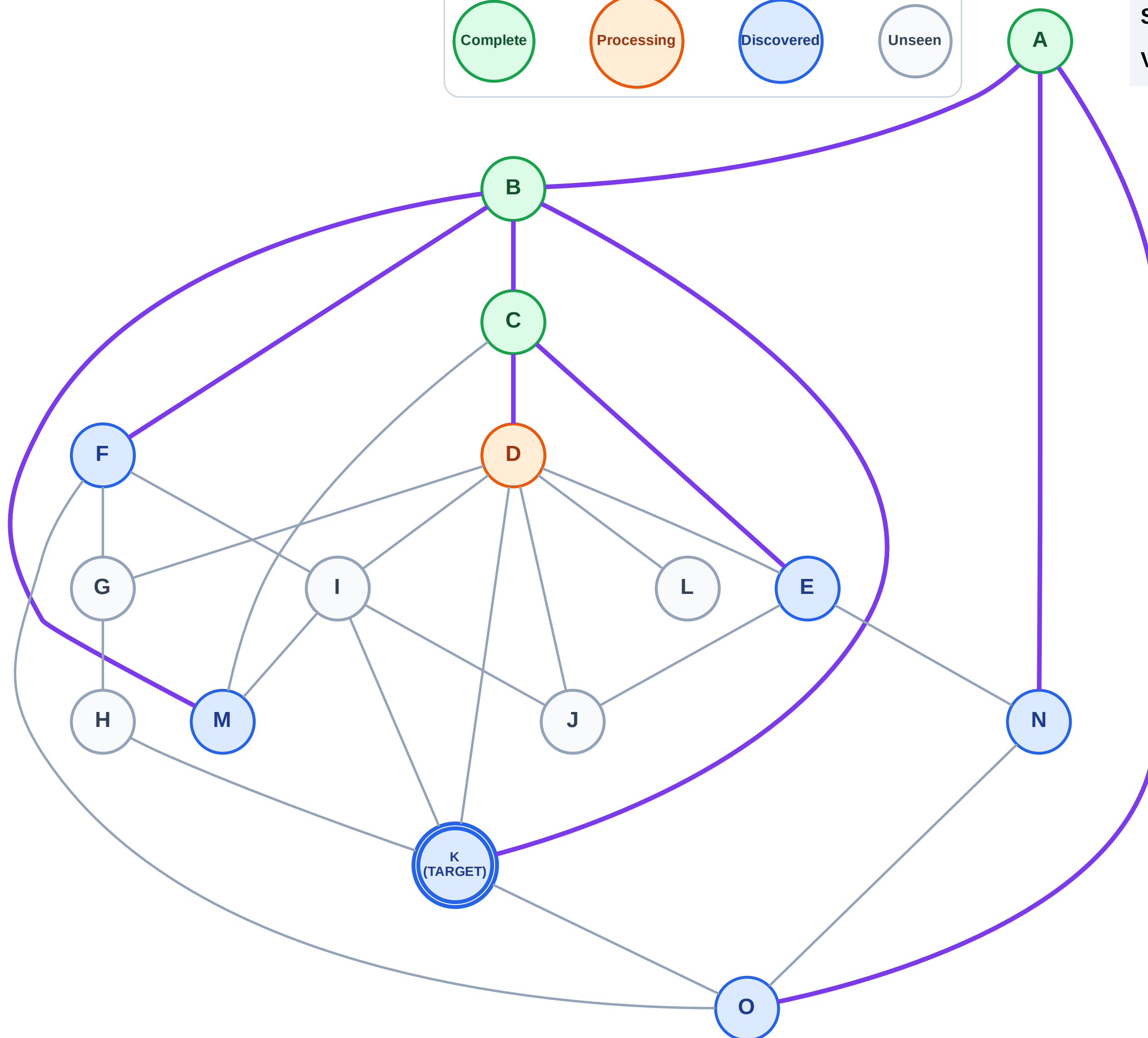
Step 8: Process node D

Legend



Stack (top at right): O | N | M | K | F | E

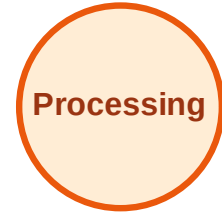
Visited: A, B, C



DFS Traversal

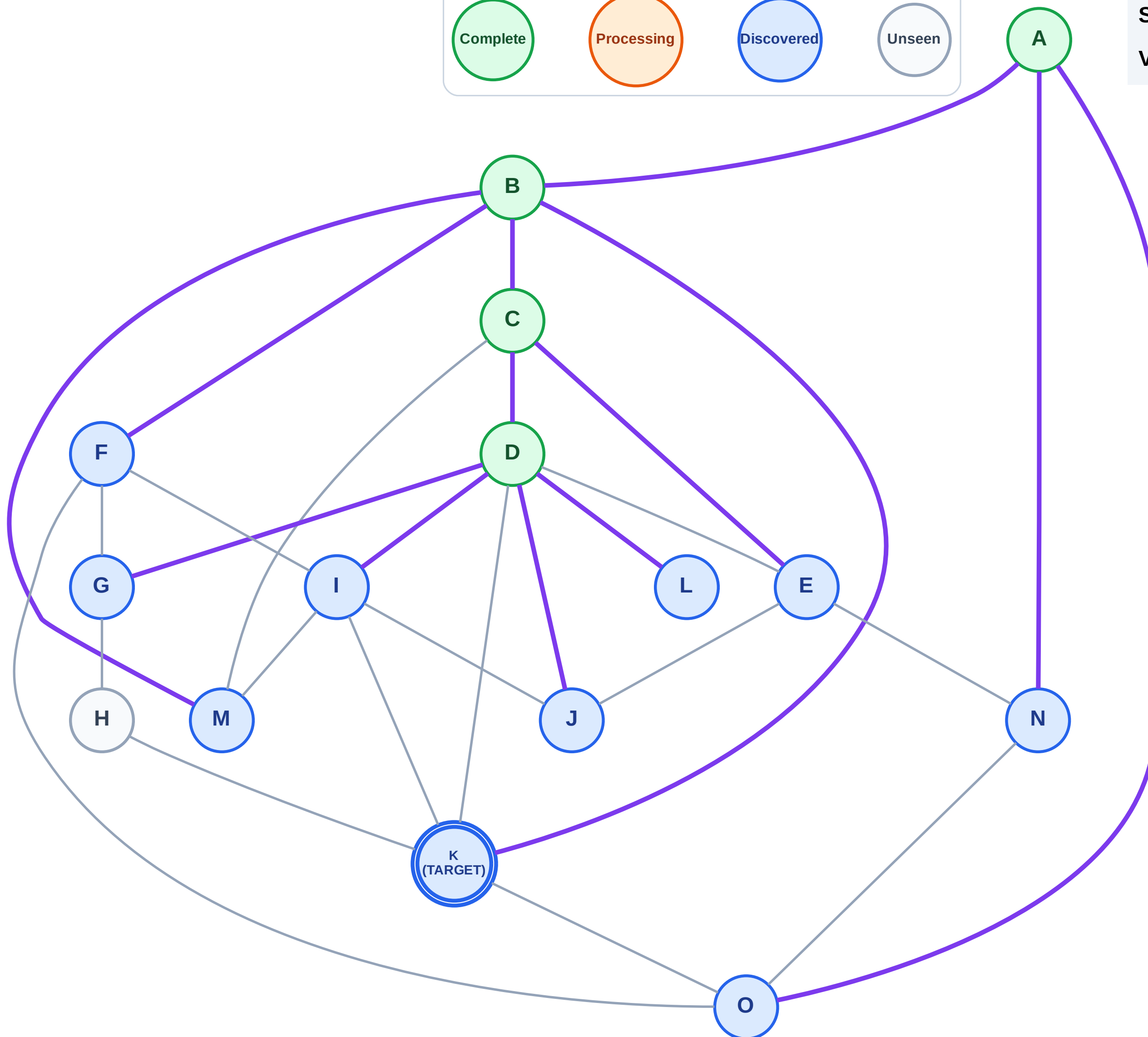
Step 9: Discover L, J, I, G from D

Legend



Stack (top at right): O | N | M | K | F | E | L | J | I | G

Visited: A, B, C, D



DFS Traversal

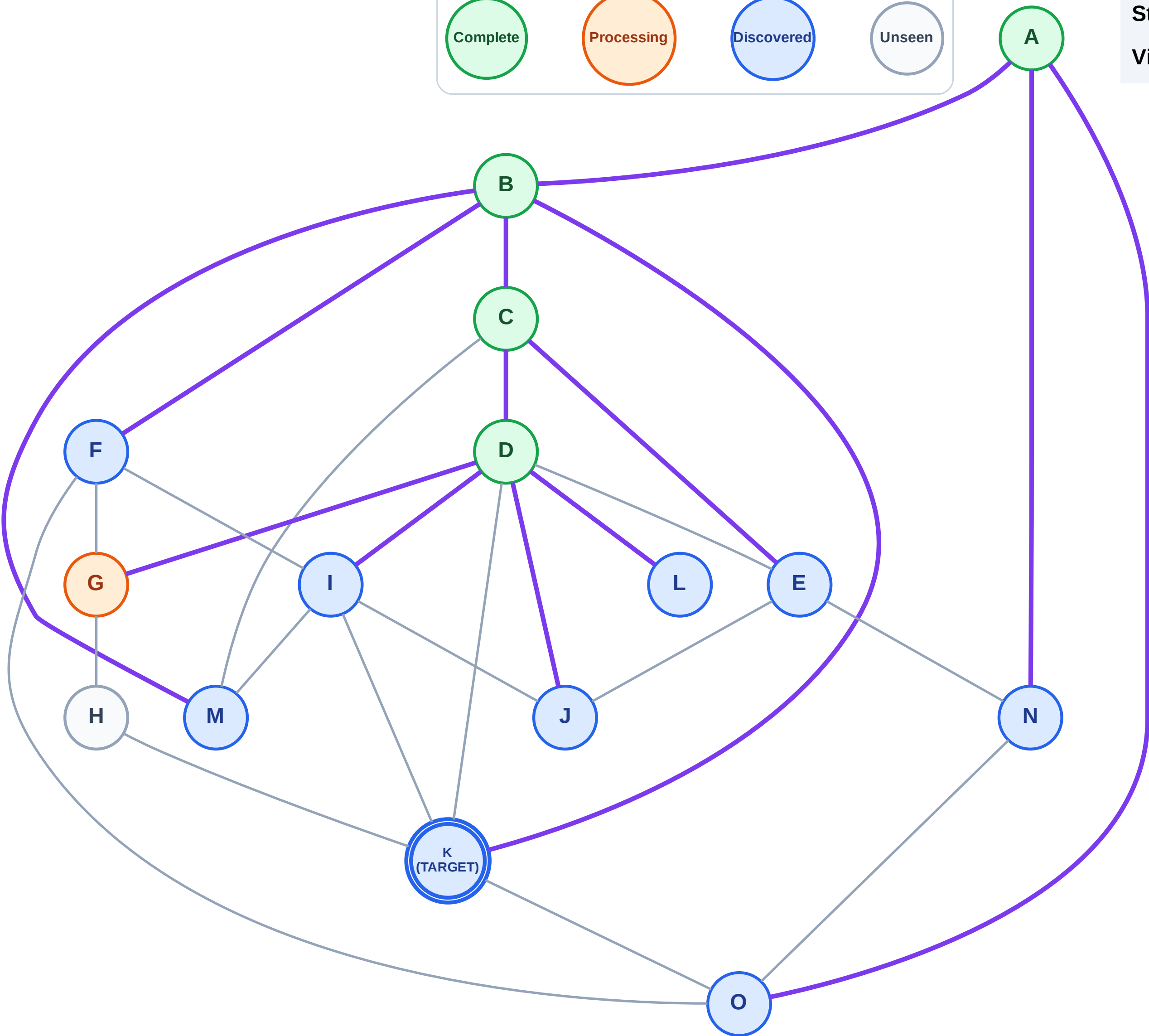
Step 10: Process node G

Legend



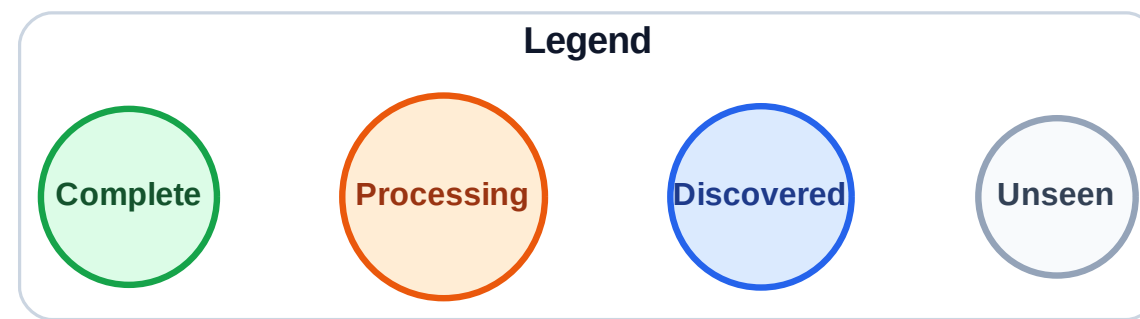
Stack (top at right): O | N | M | K | F | E | L | J | I

Visited: A, B, C, D



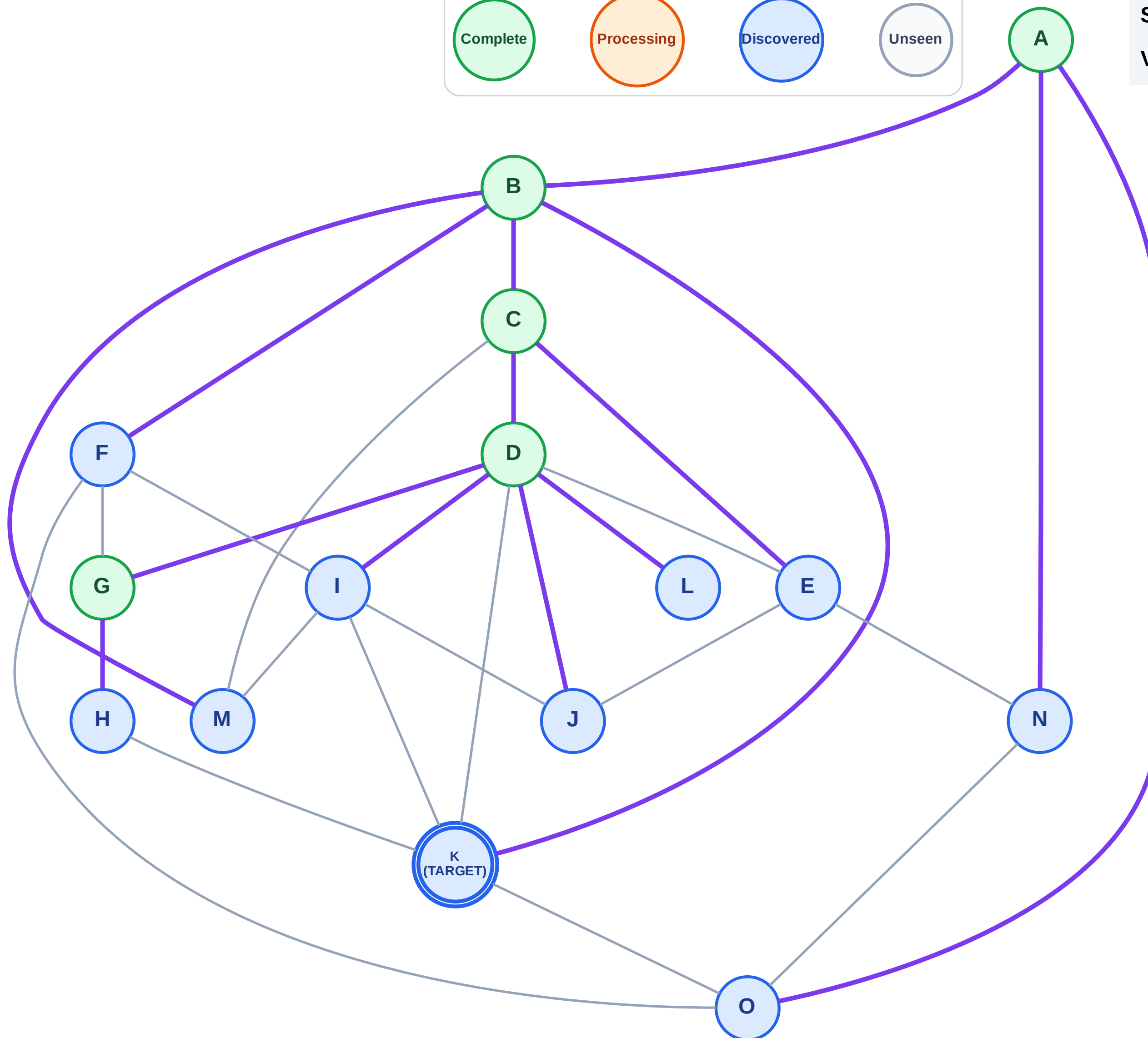
DFS Traversal

Step 11: Discover H from G



Stack (top at right): O | N | M | K | F | E | L | J | I | H

Visited: A, B, C, D, G



Step 12: Process node H

Step 12: Process node H



DFS Traversal

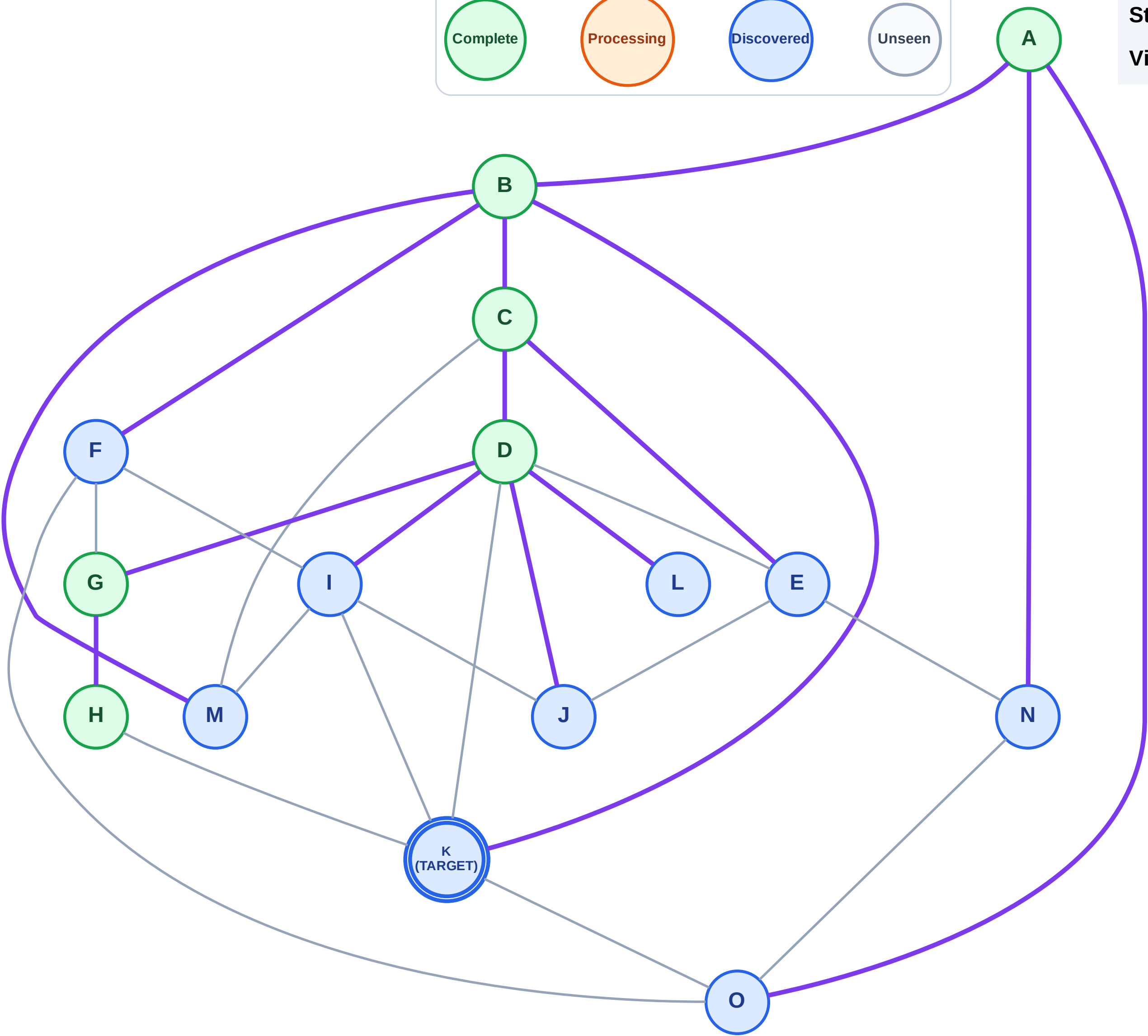
Step 13: No new neighbors from H

Legend



Stack (top at right): O | N | M | K | F | E | L | J | I

Visited: A, B, C, D, G, H



Step 14: Process node I

Step 14: Process node I

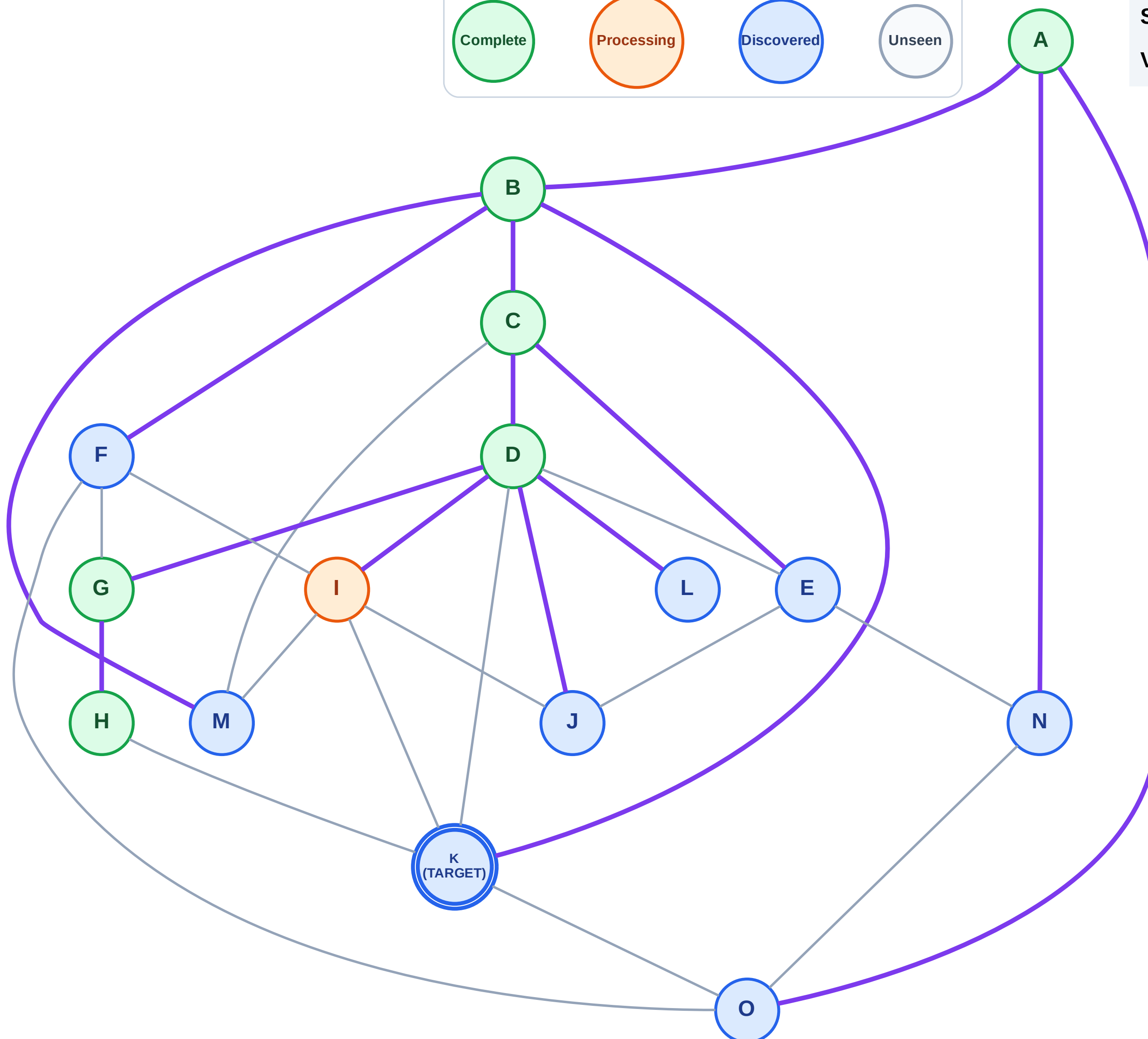
Legend



Complete Processing Discovered Unseen

Unseen

Visited: A, B, C, D, G, H



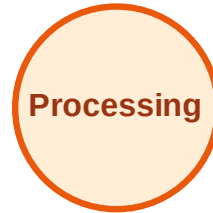
DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



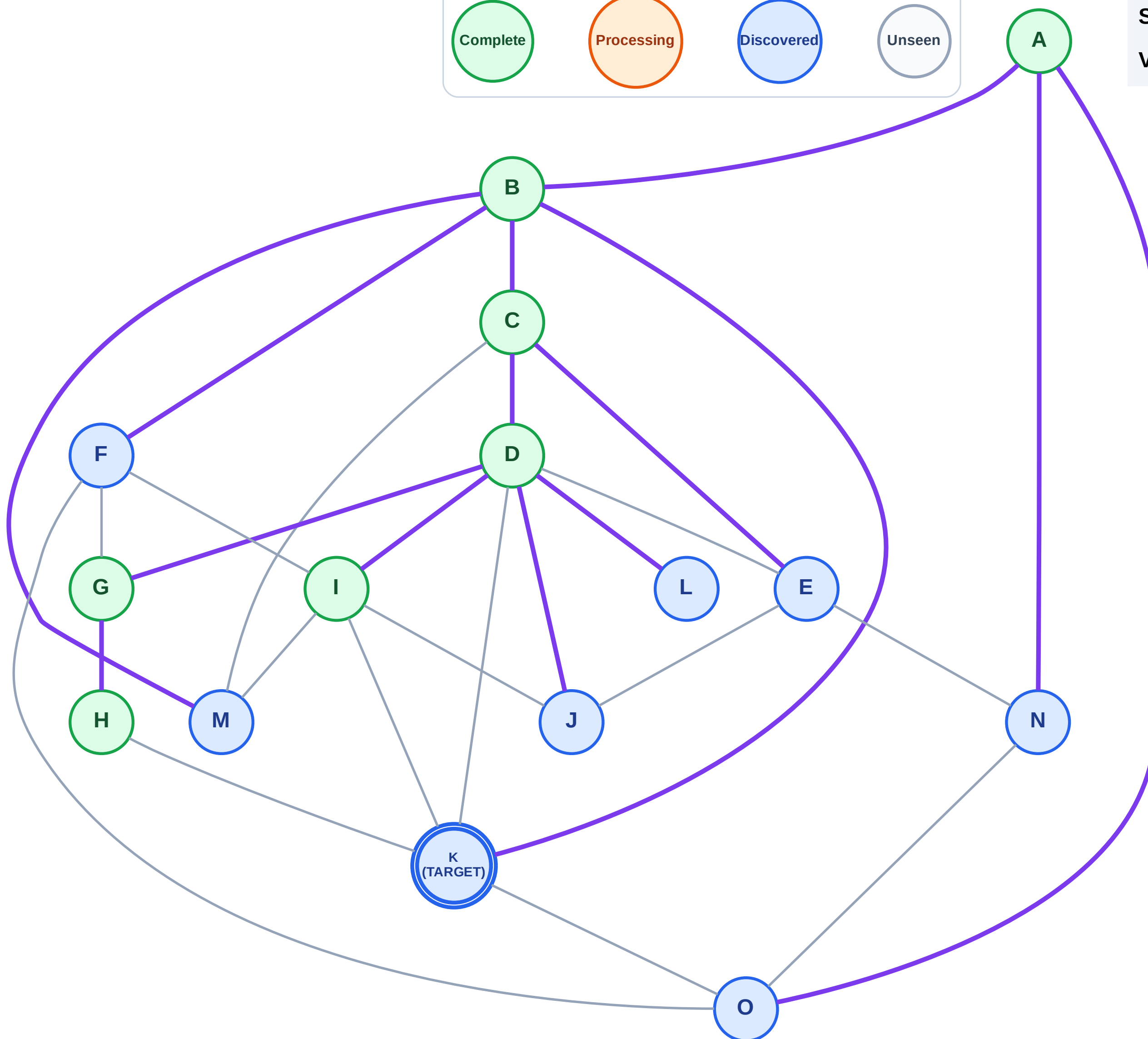
Discovered



Unseen

Stack (top at right): O | N | M | K | F | E | L | J

Visited: A, B, C, D, G, H, I



Step 16: Process node J

Step 16: Process node J

Legend

Complete

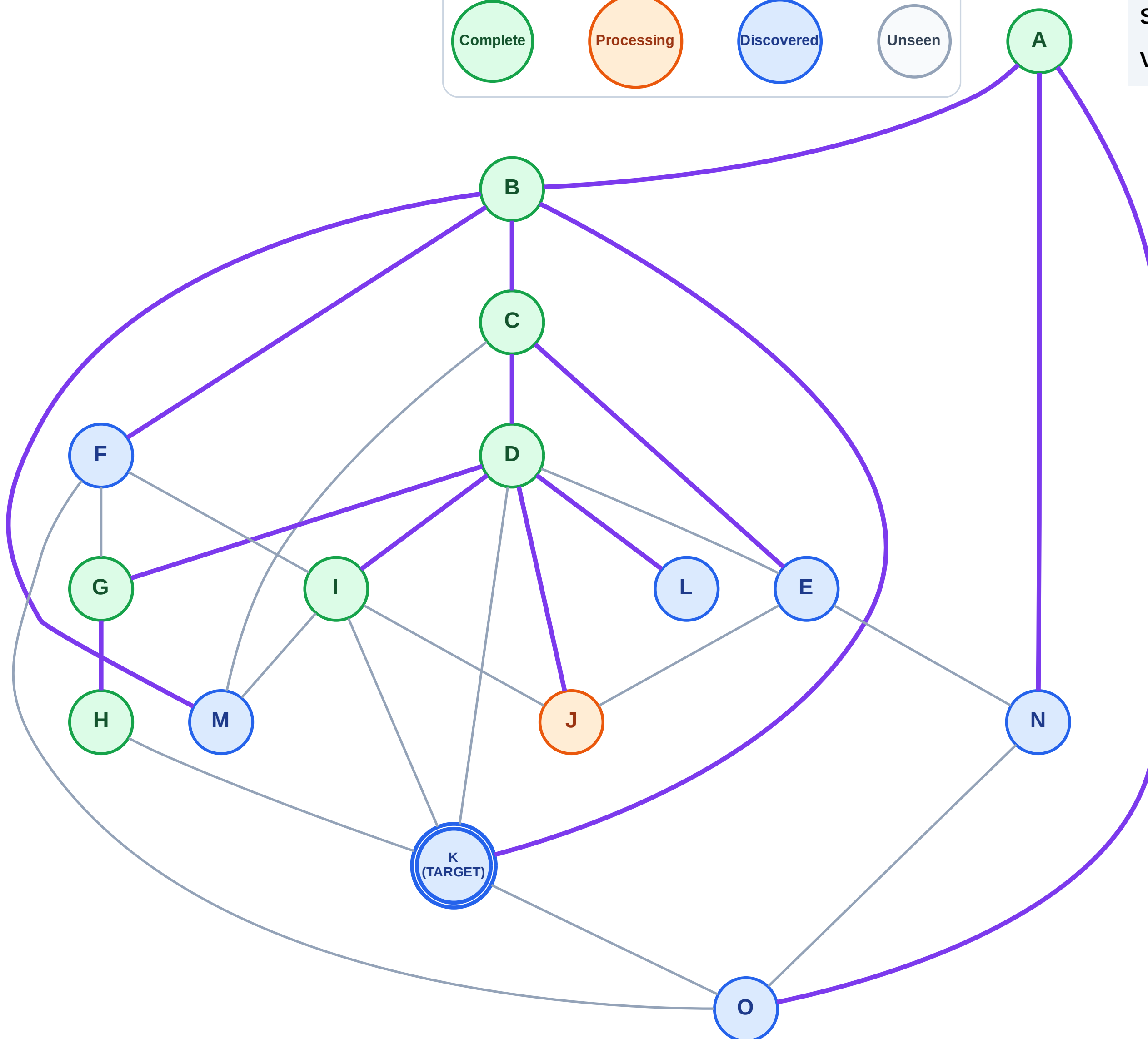
Processing

Discovered

Unseen

Stack (top at right): O | N | M | K | F | E | L

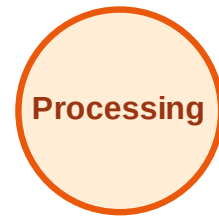
Visited: A, B, C, D, G, H, I



DFS Traversal

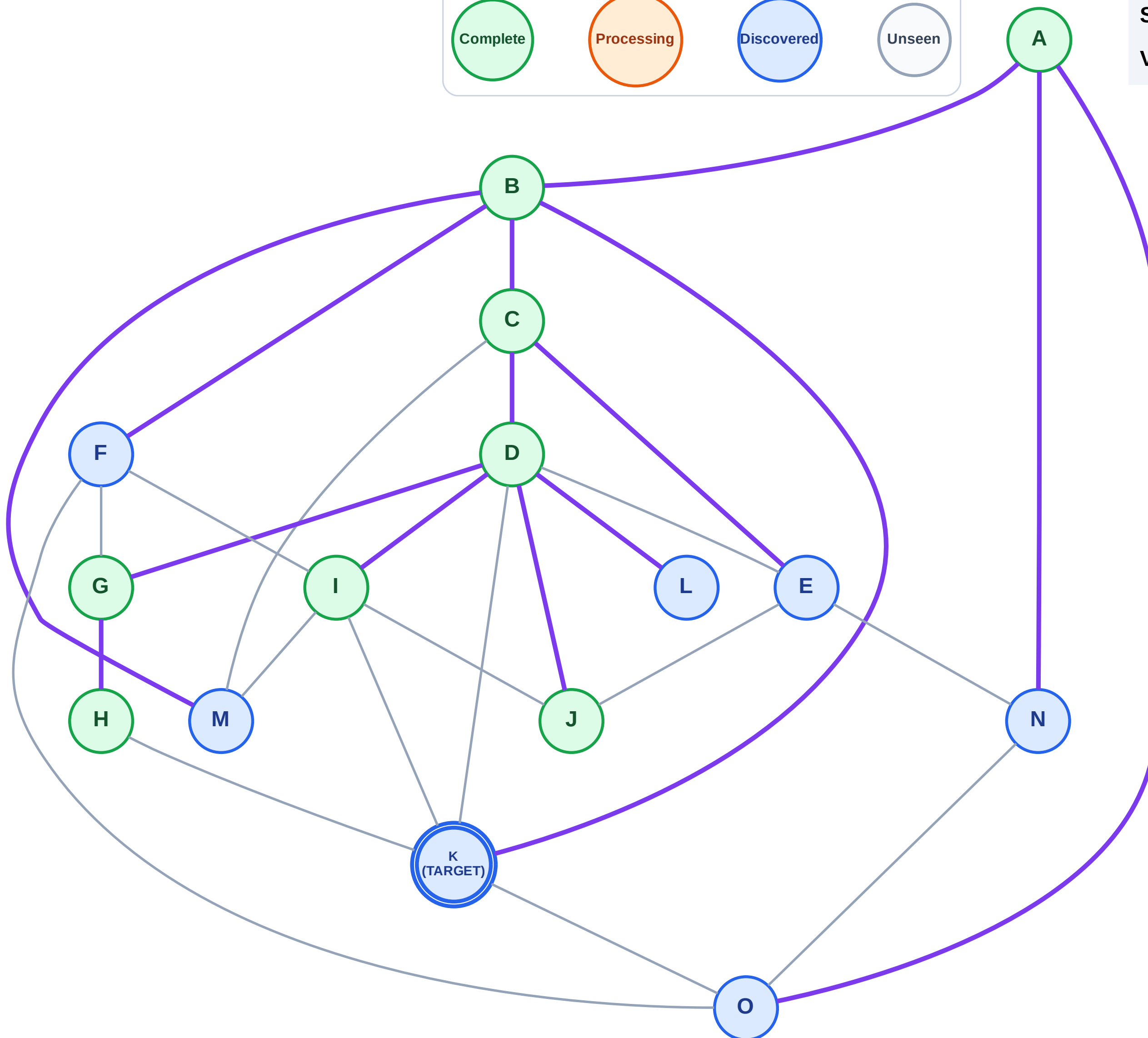
Step 17: No new neighbors from J

Legend



Stack (top at right): O | N | M | K | F | E | L

Visited: A, B, C, D, G, H, I, J



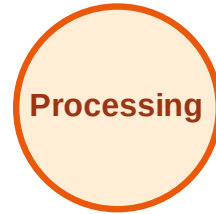
DFS Traversal

Step 18: Process node L

Legend



Complete



Processing



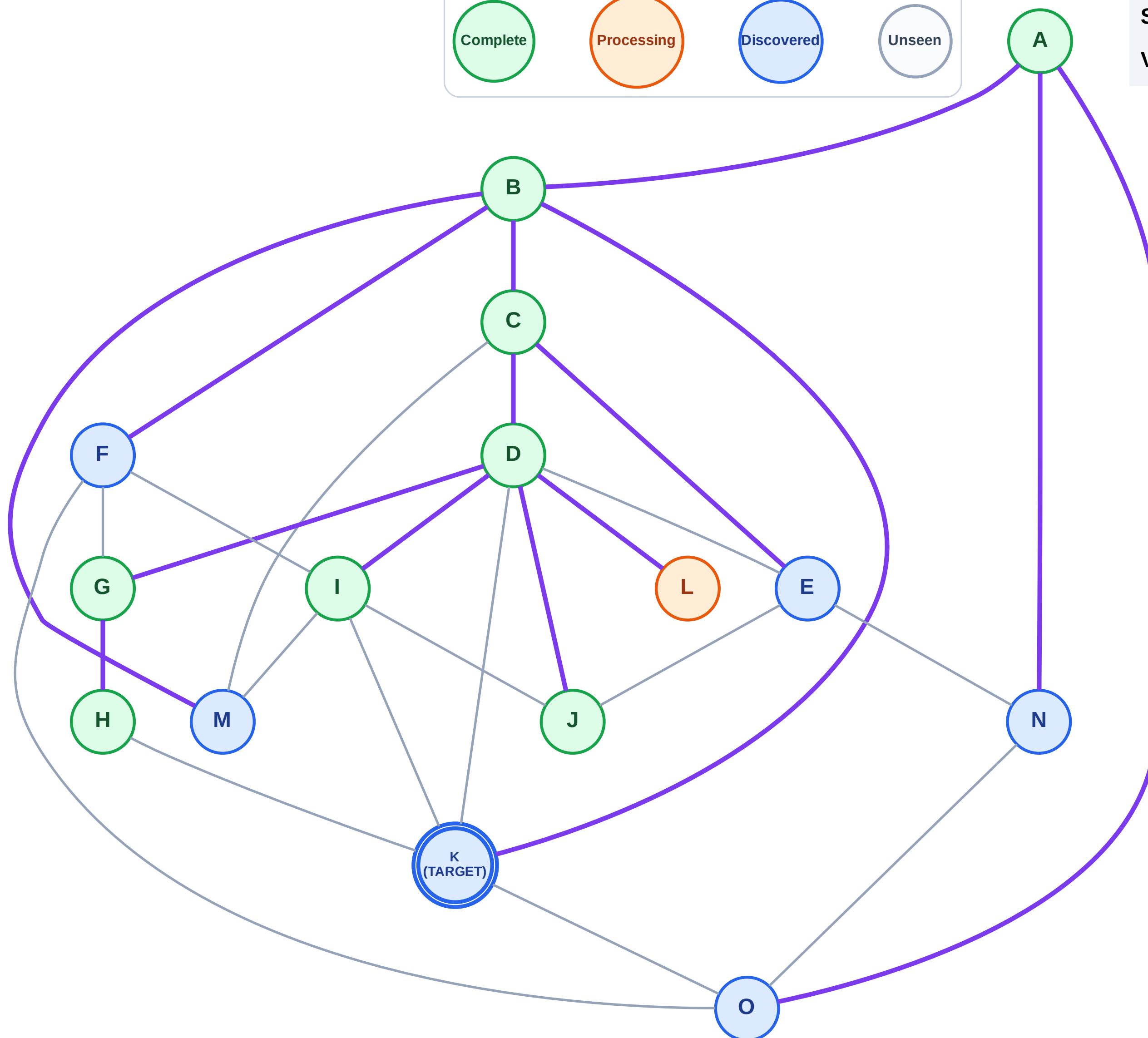
Discovered



Unseen

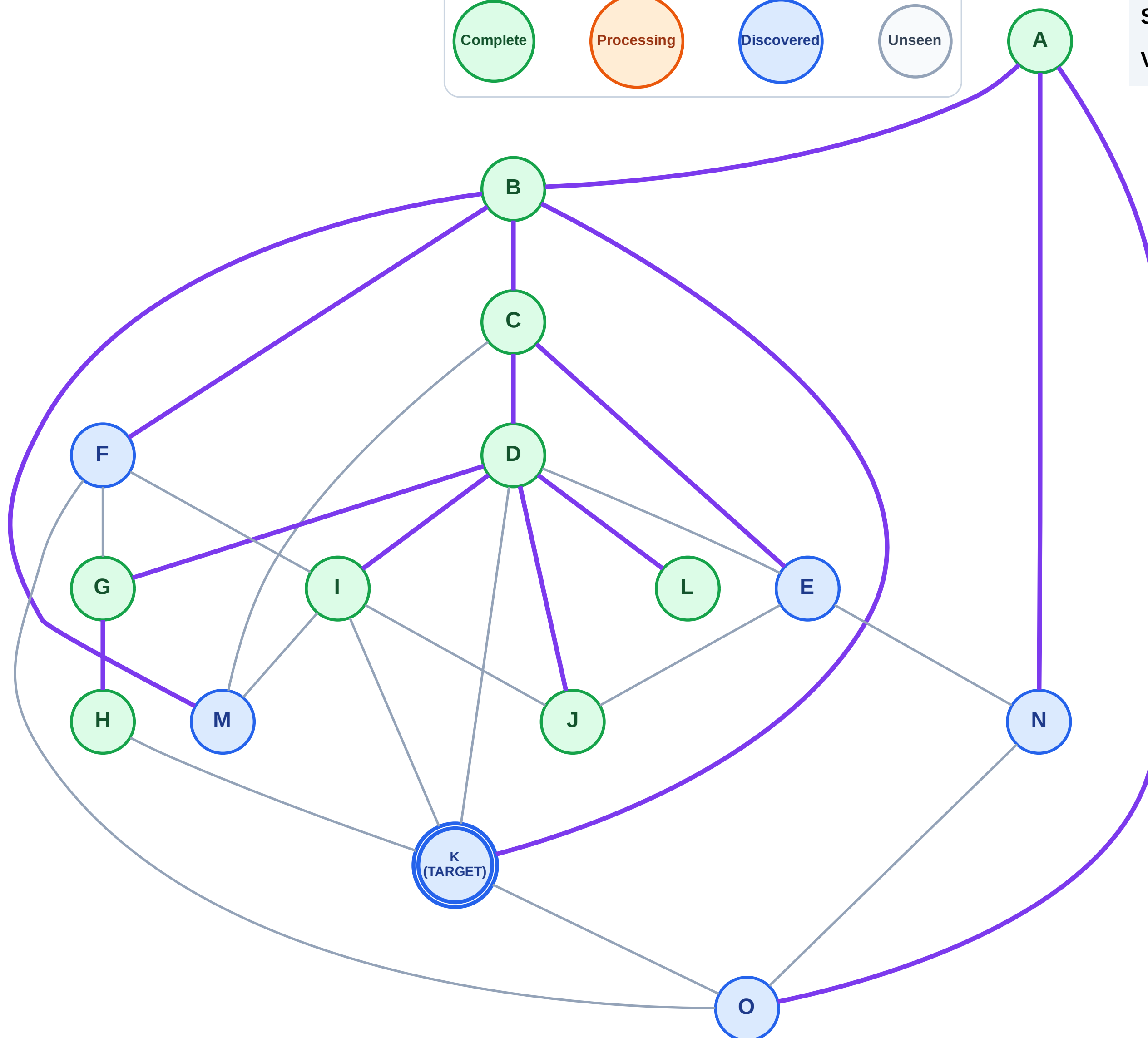
Stack (top at right): O | N | M | K | F | E

Visited: A, B, C, D, G, H, I, J



Step 19: No new neighbors from L

Step 19: No new neighbors from L



DFS Traversal

Step 20: Process node E

Legend



Complete



Processing



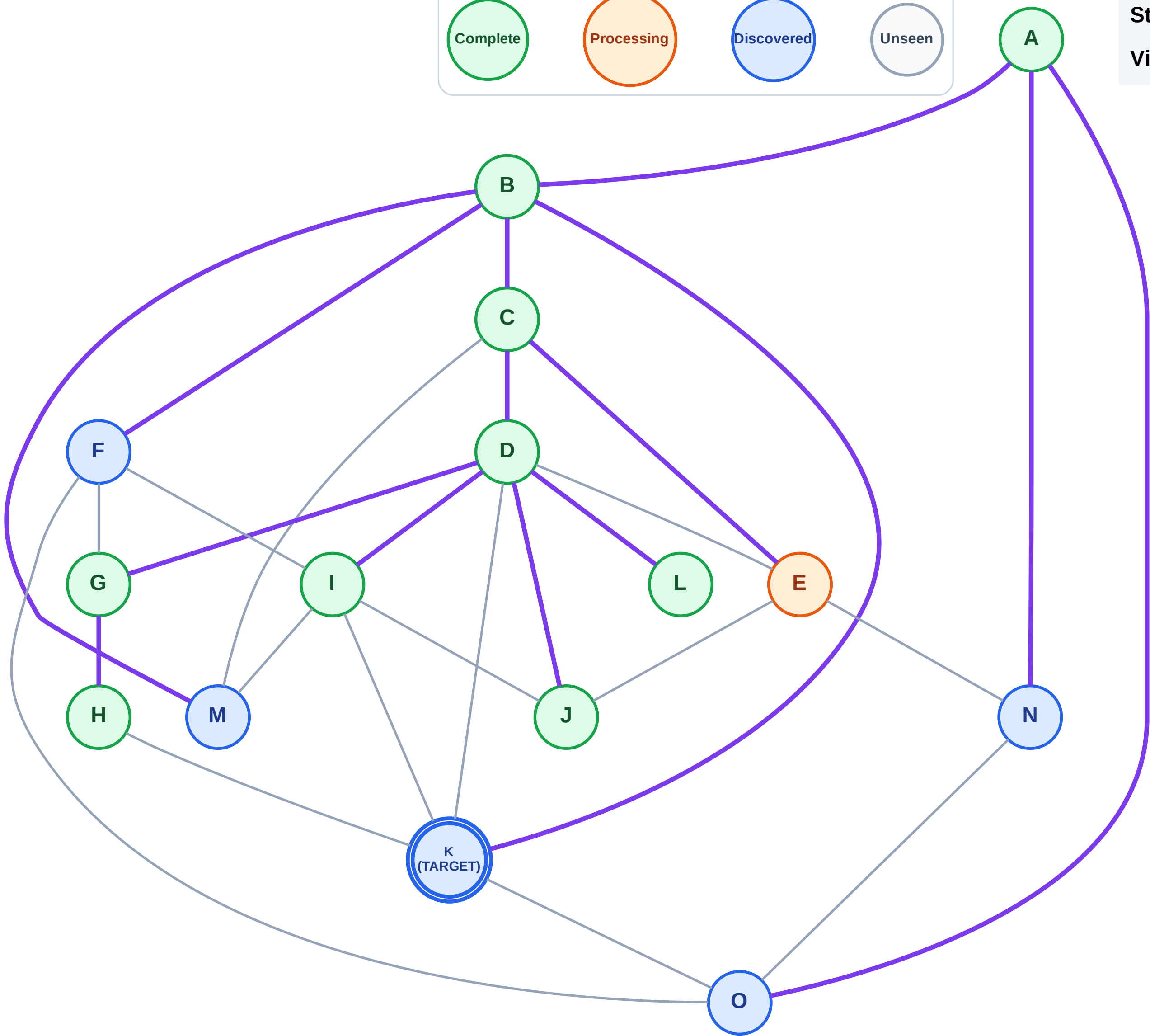
Discovered



Unseen

Stack (top at right): O | N | M | K | F

Visited: A, B, C, D, G, H, I, J, L



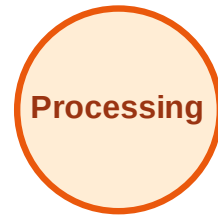
DFS Traversal

Step 21: No new neighbors from E

Legend



Complete



Processing



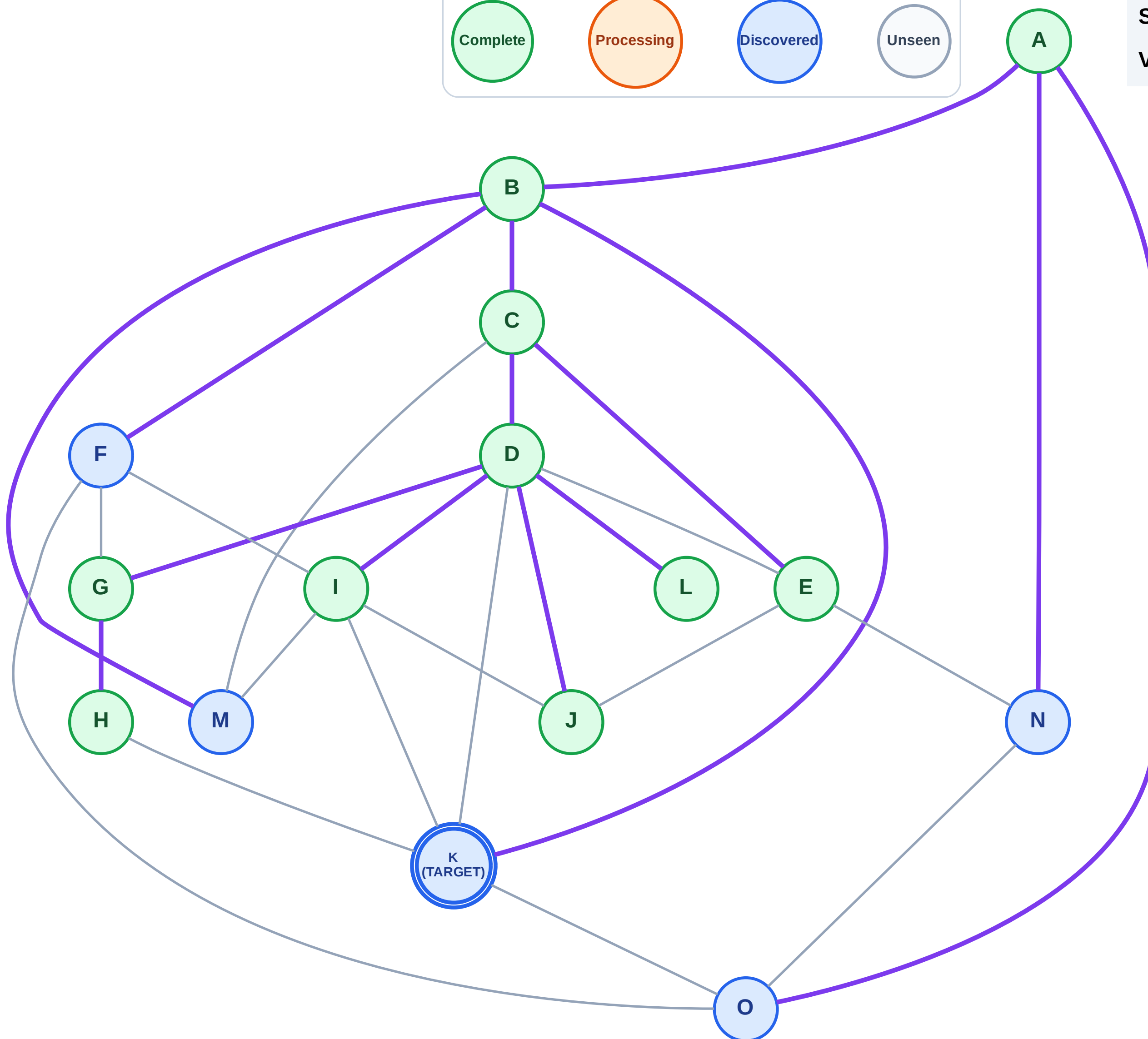
Discovered



Unseen

Stack (top at right): O | N | M | K | F

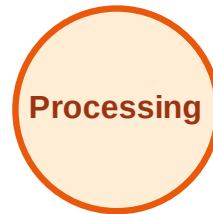
Visited: A, B, C, D, E, G, H, I, J, L



DFS Traversal

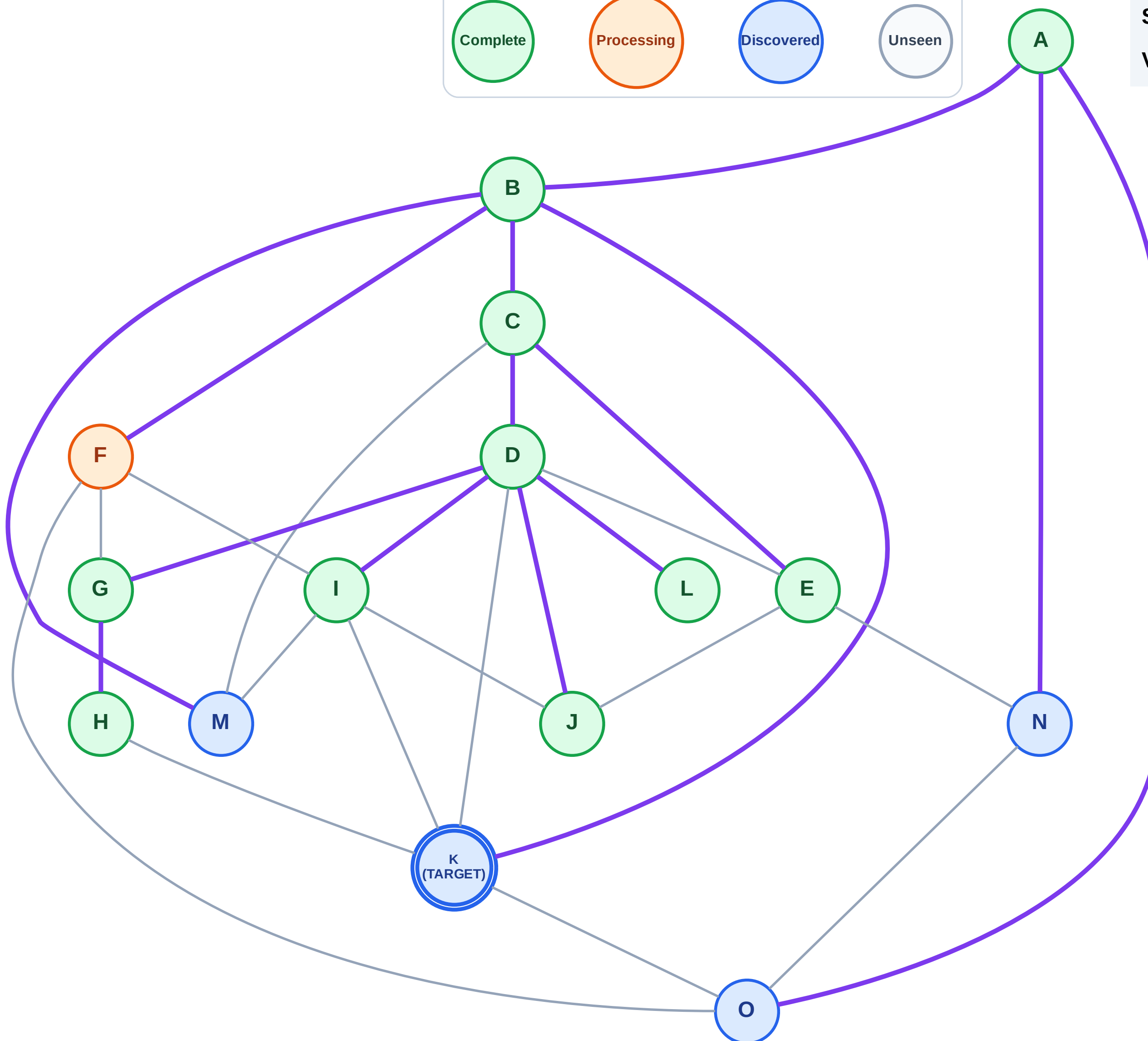
Step 22: Process node F

Legend



Stack (top at right): O | N | M | K

Visited: A, B, C, D, E, G, H, I, J, L



DFS Traversal

Step 23: No new neighbors from F

Legend

Complete

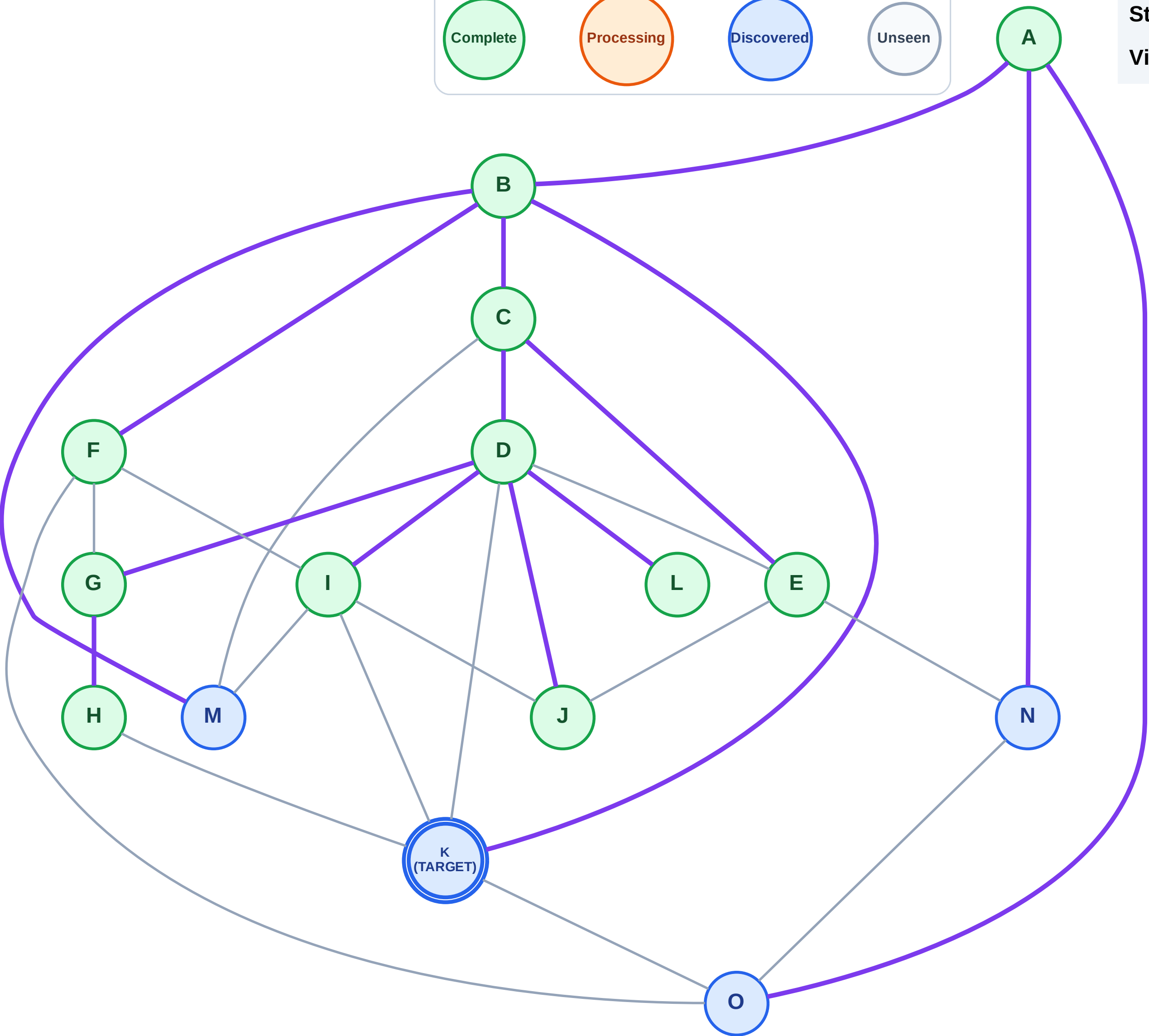
Processing

Discovered

Unseen

Stack (top at right): O | N | M | K

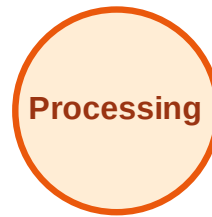
Visited: A, B, C, D, E, F, G, H, I, J, L



DFS Traversal

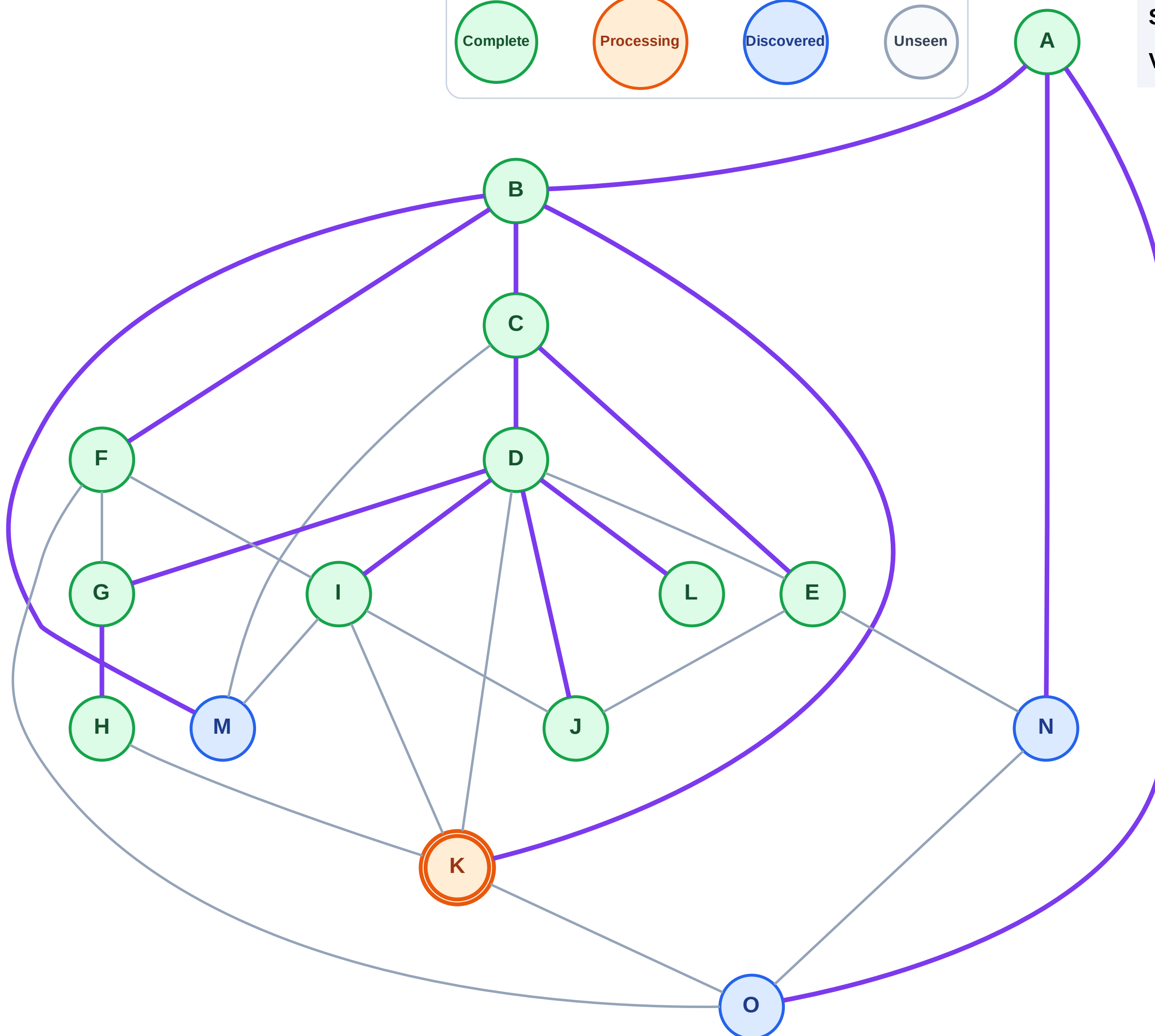
Step 24: Process node K

Legend



Stack (top at right): O | N | M

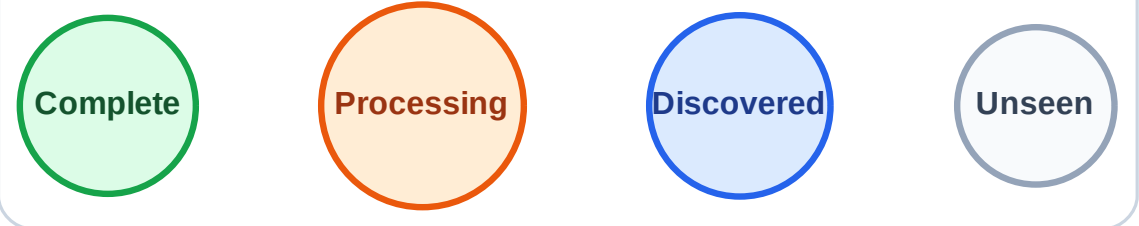
Visited: A, B, C, D, E, F, G, H, I, J, L



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): O | N | M
Visited: A, B, C, D, E, F, G, H, I, J, K, L

